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Microbial warfare against nematodes: A review of nematicidal compounds for horticulture, environment, and biotechnology

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ABSTRACT

Plant-parasitic nematodes (PPNs) pose a significant threat to global agriculture, with annual crop losses estimated at $157 billion. Although synthetic nematicides are effective, their extensive use raises concerns for environmental and human health, highlighting the need for safer, sustainable alternatives in horticulture and environmental biotechnology. This review delves into microbial strategies against PPNs, focusing on biocontrol agents (BCAs) such as bacteria and fungi that produce nematicidal compounds, including volatile organic compounds (VOCs), secondary metabolites, toxins, and extracellular enzymes. These BCAs, often attracted by plant root exudates, interact within the rhizosphere to suppress nematodes, offering promising, eco-friendly control methods for sustainable agriculture. Advances in multi-omics techniques-proteomics, transcriptomics, metabolomics, and metagenomics-have accelerated the discovery of novel nematicidal compounds, paving the way for new applications in biocontrol. Notably, nematophagous fungi (NF), present in diverse taxa such as Ascomycota and Basidiomycota, exhibit significant potential through direct (e.g., parasitism, toxin production) and indirect (e.g., lifecycle disruption) effects on nematodes. This review underscores the ecological roles, taxonomy, and mechanisms of NF in sustainable horticulture and agriculture and advocates for the integration of microbial biocontrol within an integrated pest management (IPM) framework. Such an approach is essential to reduce reliance on synthetic pesticides, enhance nematode control, and promote resilient agroecosystems.

1. Introduction

Nematodes in soil ecosystems serve diverse functions, with some species promoting soil health, while others can be harmful. These organisms, small and worm-like in nature, belong to the phylum Nematoda and mainly inhabit terrestrial environments. Their influence on soil processes is significant, particularly in nitrogen cycling and biomass distribution in plants (Fig. 1), as demonstrated through controlled experiments. Studies using Petri dishes have shown that in the presence of bacterivorous and fungivorous nematodes, a greater amount of nitrogen is released as ammonium, compared to environments without these nematodes (Girgan et al., 2020). The nitrogen, produced through microbial grazing, then become available to plants and has a well-documented effect on plant biomass distribution. Plant-parasitic nematodes (PPNs), or the phytonematodes, are obligate invertebrate parasites affecting a vast range of plants. Till now there are about 197 genera comprising 4300 species of the phytonematodes, with notable ones being *Meloidogyne* (root-knot nematodes), *Heterodera* and *Globodera* (cyst nematodes), *Pratylenchus* (lesion nematodes), *Xiphinema* (dagger nematode), *Tylenchulus* (citrus nematode), *Radopholus* (burrowing nematode), *Helicotylenchus* (spiral nematode), *Belonolamius* (sting nematode) and *Rotylenchulus* (reniform nematode) (Abd-Elgawad et al., 2016). These PPNs present a serious global threat to agriculture, contributing to an estimated 12.6 % yield loss, valued at $157 billion, across about 20 main crops like tomato, rice, soybean and peanut (Ning et al., 2022). The Root-Knot Nematodes (RKNs) like *Meloidogyne javanica*, *Meloidogyne arenaria, Meloidogyne hapla* and *Meloidogyne incognita* are among the most destructive, causing up to 90 % crop damage and making plants more vulnerable to other pathogens by hindering water and nutrient uptake (Migunova and Sasanelli, 2021) (Fig. 1). PPNs weaken plant defenses by secreting effector proteins that interfere with the host's defense mechanisms, further exacerbating their economic impact on agriculture (Ali et al., 2023). These nematodes primarily reside in the soil and affect various plant parts, including roots, flowers, stems and leaves (Fig. 1). Their way of feeding is by penetrating in the cells of plant with their flexible stylet, connected to pharyngeal glands that assist in the penetration, delivery of effector and the parasitism. The Plant-parasitic nematodes (PPNs) are classified as ectoparasitic or

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endoparasitic based on their feeding behaviors, with sedentary endoparasites, like cyst nematodes and root-knot nematodes, being particularly damaging. The growing population necessitates the development of sustainable agricultural practices to manage PPNs more effectively (Mokrini et al., 2024). Managing nematodes is more complex than controlling other pests due to their soil-dwelling nature and preference for attacking underground plant parts. Control strategies aim to reduce nematode populations below economically harmful thresholds, with the most common method being the use of synthetic nematicides. Historically, plant-parasitic nematodes in intensive farming systems have been managed using chemical fumigants and nematicides. However, the overuse of these chemicals has raised environmental and health concerns, leading to restrictions and bans on certain compounds (Rich et al., 2004). The phase-out of methyl bromide in 2005, for example, significantly impacted nematode control strategies, and today, only a limited range of nematicides, mostly organophosphates and carbamates, are available in many countries. These chemicals, which are highly toxic and prone to contaminating groundwater, have driven the search for alternative control methods. For millennia, farmers have utilized bacteria to promote plant growth, and more recent research has highlighted the role of soil bacteria in converting atmospheric nitrogen into forms usable by plants, thereby improving soil fertility (Nannipieri et al., 2017). A wide variety of microorganisms interact with plants, especially in the rhizosphere, and these relationships can either benefit or harm plant roots. Biological control agents (BCAs) present a promising and environmentally friendly alternative for managing PPNs in a cost-effective manner (Ali et al., 2023). Certain rhizospheric fungi and bacteria like *Pseudomonas fluorescens, Bacillus thuringiensis*, *B. amyloliquefaciens, Xenorhabdus bovienii*, *Trichoderma harzianum*, *Pasteuria penetrans*, and *Paecilomyces lilacinus*, have proven effective in controlling PPNs through mechanisms like inhibiting hatching of eggs and destroying the females or the juvenile nematodes (Kishore et al., 2022).These BCAs work against PPNs either directly or indirectly by competing for the antibiotics, nutrients, enzymes, toxic metabolites or the volatile compounds. The indirect antagonism often involves inducing plant defense responses, such as Induced Systemic Resistance (ISR), which enhances a plant's resilience to pests (Migunova and Sasanelli, 2021). Fungi, which interact with nematodes either as parasites or predators, also play an essential role in controlling nematode populations. Nematophagous fungi, which are capable of capturing and consuming nematodes, are natural predators in soil ecosystems (Song et al., 2022). These fungi can detect nematodes by responding to their pheromones and trap them using specialized mycelial structures, or they may target nematode eggs and cysts, penetrating and digesting them. Nematophagous fungi vary in their saprophytic and parasitic abilities, with some species capable of growing saprophytically in soil, while others rely primarily on nematodes for sustenance. These fungi are widely distributed in soils, including organic substrates like dung. Despite preferring organic soils, their low nutrient requirements allow them to thrive in various soil types, where they help regulate nematode populations (Migunova and Sasanelli, 2021). As biological control agents, these fungi are vital in both agriculture and animal husbandry, offering a natural method to keep nematode numbers in check. In the era of advanced technologies like metabolomics, new approaches are emerging for developing nematicides that can enhance biocontrol strategies.

2. Nematodes in nature/nematodes

Nematodes, also known as roundworms or eelworms, belong to the phylum Nematoda, superphylum Ecdysozoa, and kingdom Animalia. They are among the most significant and abundant soil invertebrates, playing a crucial role at all trophic levels within the soil food web (Van Den Hoogen et al., 2019). Nematodes are commonly found in decomposed organic matter, soil, plant roots, and other organic-rich environments (Senthilkumar et al., 2021). Apart from terrestrial habitats, nematodes have adapted to various ecosystems, including freshwater (Pérez-García et al., 2018), marine environments (Gonzalez et al., 2021), and even extreme conditions like the polar regions and high-temperature environments (Sapir, 2021). It is estimated that nematodes account for four out of every five organisms on Earth, making them a subject of interest in applied ecology. Soil nematode populations

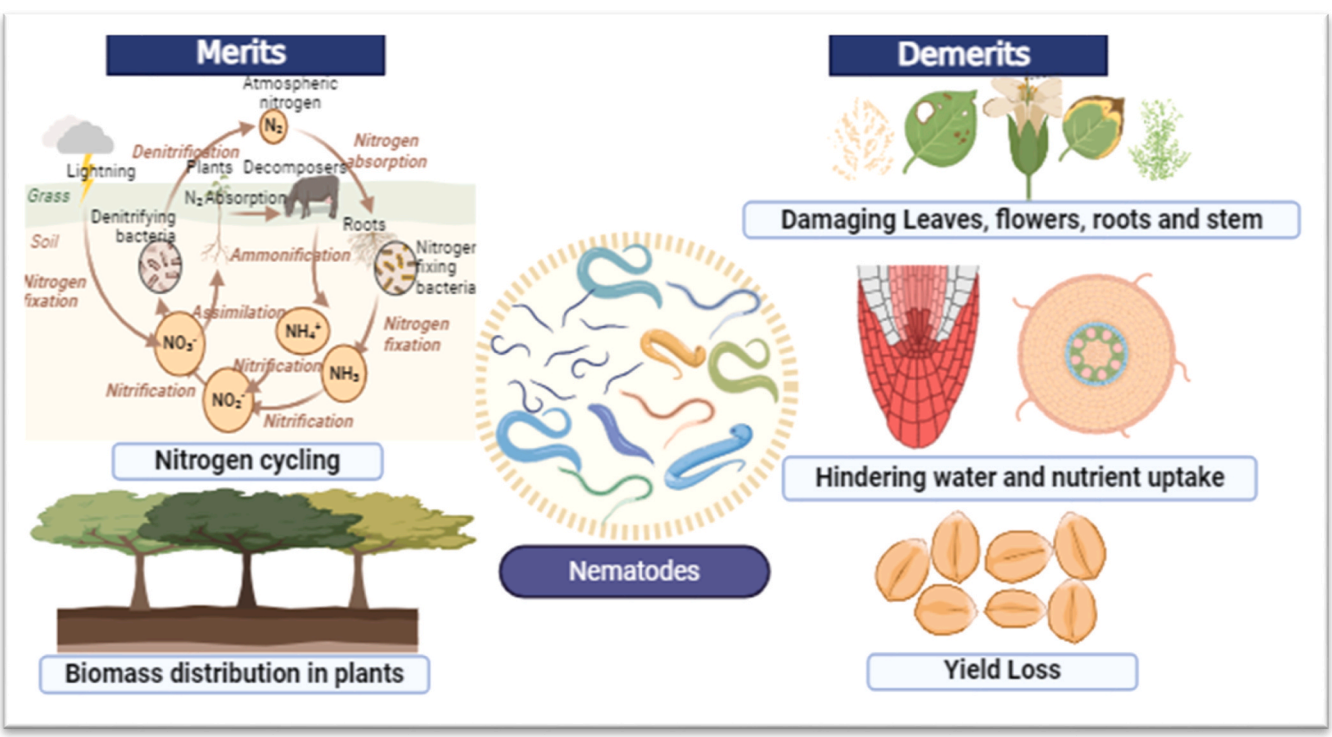

Fig. 1. Merits and demerits of Nematodes present in soil.

range from 1 to 100 million individuals per square meter of soil, predominantly in the upper layers, where they live in water films and pore spaces. Certain species, such as *Caenorhabditis elegans*, have become model organisms for genetic research, with the number of described species increasing from 15,000 (Bongers et al., 1997) to over 25,000 (Geisen et al., 2019) in just two decades. Nematodes are now one of the best-characterized taxonomic groups in soil ecosystems. They interact with a wide range of microorganisms and exhibit various ecological relationships, such as parasitism and predation (van Rensburg et al., 2021). Additionally, nematodes play a vital role in the food chain by serving as prey for organisms like mites (Walter et al., 1986) and other nematodes (Aguilar-Marcelino et al., 2020), while also feeding on fungi, bacteria, and microarthropods (Ferris, 2010). Moreover, they contribute to biogeochemical cycling and promote vegetation dynamics (Van Den Hoogen et al., 2019). Nematodes undergo a six-stage life cycle: the egg, four juvenile stages (J1–J4), and the adult stage. Juveniles resemble adults but are smaller and lack fully developed gonads, which form during the final molt (Jonathan, 2022). The structure of nematode communities serves as a useful indicator for assessing soil quality and function due to their presence in decomposing organic matter, diverse interactions with other soil organisms, and rapid response to environmental changes. They are easily isolated from soil samples, and genus identification is relatively straightforward (Bongers and Bongers, 1998). Soil samples contain a wealth of nematodes, which provide valuable ecological information (Yeates et al., 1993; Bongers and Bongers, 1998; Bongers and Ferris, 1999). Their feeding habits and oral structures make it easy to classify them into functional groups. Nematodes not only feed on plant roots but also on bacteria, fungi, and even other nematodes, with changes in their abundance reflecting shifts in the soil's decomposition processes. Nematodes are widely used as bioindicators of soil health and quality. The development of the Maturity Index (MI) (Bongers, 1990) and its modifications (Yeates, 1994), along with other nematode-based indices (Ferris et al., 2001; Ferris, 2010), has greatly advanced the study of soil ecosystems. The MI evaluates environmental disturbances by analyzing the nematode community structure, while the Plant Parasite Index (PPI) focuses specifically on plant-parasitic nematodes and their distinct responses to environmental changes (Bongers, 1990; Bongers et al., 1997). A modification of the MI by Yeates (1994) (Yeates, 1994) combined free-living and plant-parasitic nematodes to provide a more comprehensive overview of soil conditions. Combining the MI and PPI resulted in the Summarized Maturity Index (Σ MI). Later, Bongers et al. (1997) proposed excluding pollution-sensitive free-living nematodes (c-p 1) from the MI, leading to the development of the Maturity Index 2–5 (MI 2–5), which is particularly useful for monitoring soil affected by pollutants like heavy metals. Nematodes have evolved over millions of years to adapt to various biological systems, becoming parasites of plants, animals, and humans (Blaxter and Koutsovoulos, 2015). In agricultural systems, soil nematodes are categorized into three groups: (1) entomopathogenic nematodes that feed on insects, (2) free-living nematodes that consume microorganisms like bacteria and fungi, and (3) plant-parasitic nematodes that feed on plant tissues (Grabau and Dickson, 2021).

3. Ecological functions of nematodes

Nematodes fulfil several key ecological functions that contribute significantly to ecosystem dynamics. Some of these functions include (Fig. 2):

- Decomposition of Organic matter: Nematodes play a significant role by breaking down organic matter in the soil by feeding on dead plant material and animal remains. This process helps fragment organic matter, making it easier for further decomposition and promoting nutrient recycling (Ferris et al., 2001).
- Predation and pest control: Certain nematodes act as predators, feeding on bacteria, fungi, insects, and plant-parasitic nematodes. Predatory nematodes from the orders Mononchida, Dorylaimida, Diplogasterida, and Aphelenchida are particularly important for managing soil pest populations through their varied mouth structures, feeding strategies, and dietary preferences (Khan and Kim, 2007).
- Soil bioturbation: Nematodes contribute to soil mixing and loosening as they move through it. This bioturbation improves soil structure, enhances gas exchange, promotes water movement, facilitates

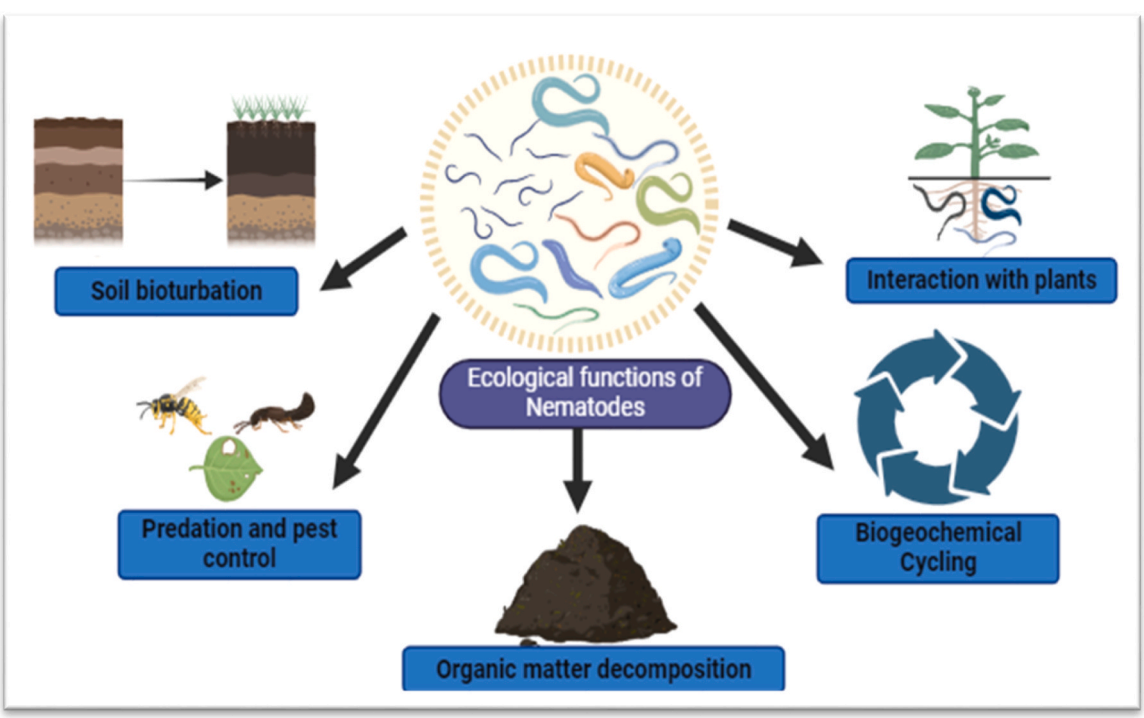

Fig. 2. Ecological functions of Nematodes.

nutrient availability to plant roots, and physically alters the soil (Gabet et al., 2003).

- Biogeochemical cycling: Nematodes influence key nutrient cycles, such as those of nitrogen, phosphorus, and carbon, through their interactions with microorganisms and plants. Their feeding, secretions, and activities affect nutrient availability and distribution within ecosystems (Ferris et al., 2001; Schmidt et al., 2007).
- Interactions with plants and other organisms: Nematodes engage in both mutualistic (Wood et al., 2018) and parasitic relationships with plants (Pulavarty et al., 2021). Some species assist in nitrogen fixation in collaboration with certain plants, while others may cause harm by damaging plant roots or spreading plant diseases (Wood et al., 2018; Pulavarty et al., 2021).

4. Plant-parasitic nematodes (PPNs): general categories and the management strategies of PPNs

Plant-parasitic nematodes are small, worm-like pathogens, typically less than 1 mm in length, that feed on plant tissues (Silva and Uchida, 2000). They can infest a wide variety of host plants, including ornamental plants, agronomic and vegetable crops, fruit and nut trees, turfgrass, and forest trees (Williams et al., 2021). These unsegmented nematodes have cylindrical, thread-like bodies that taper at both ends. Female nematodes vary in shape, ranging from cylindrical to pear-shaped (Fourie et al., 2017). A characteristic feature of most plant-parasitic nematodes is a needle-like structure known as a "stylet" located at the mouth, which they use to pierce and consume plant tissues, especially root cells, as seen in the destructive root-knot nematodes.

Above-ground nematode parasites are less common and include stem and bulb nematodes, seed gall nematodes, and foliar nematodes. Based on their feeding behavior, plant-parasitic nematodes are classified into two general groups: (1) ectoparasites, which feed on the outer root tissues like the epidermis and cortical cells without entering the roots, and (2) endoparasites, which penetrate the roots and feed on the internal cells (Fig. 3) (Ankrom et al., 2020).

Subterranean plant-parasitic nematodes can be categorized into four main types based on their parasitic mode:

(i) **Ectoparasites** such as *Helicotylenchus* spp. and other spiral nematodes, which feed on the root apex and peripheral cells by inserting their long, robust stylets without entering the roots. They typically move through the soil in search of plant roots to parasitize.

(ii) **Migratory endoparasites**, including species like *Pratylenchus* spp. and *Ditylenchus dipsaci*, can penetrate plant roots, move within them while feeding, and exit to find new roots. Both ectoparasites and migratory endoparasites go through developmental stages (J3, J4, and adulthood) without noticeable sexual dimorphism.

(iii) **Semi-endoparasites**, such as *Tylenchulus semipenetrans* and *Rotylenchulus reniformis*, penetrate roots with only their head, while their posterior remains in the soil. These nematodes remain stationary, with the female body often swelling outside the root.

(iv) **Sedentary endoparasites**, including root-knot nematodes (*Meloidogyne* spp.) and cyst nematodes (*Heterodera* and *Globodera* spp.), enter the roots and establish themselves inside.

Semi-endoparasites and sedentary endoparasites exhibit sexual dimorphism, with the swollen females often protruding from the root surface. Migratory and sedentary endoparasites cause damage to plant tissues during their invasion, movement, and feeding on host plants. Most research on plant-nematode interactions has focused on root-knot and cyst nematodes, as these species are widespread and lead to significant agricultural losses globally. Plant-parasitic nematodes (PPNs) interact with their hosts in different ways, ranging from transient ectoparasites to more intimate relationships like those formed by sedentary endoparasites (Table 1).

4.1. Life cycle of nematode

The life cycle of most plant-parasitic nematodes is a complex process, involving several stages, including both free-living stages in the soil and parasitic stages within host roots. This life cycle can be simplified into two main phases: pre-parasitic and parasitic, with six stages in total-egg, four juvenile stages, and the adult stage. The first four stages are immature and referred to as juvenile stages, and the overall life cycle is relatively direct and straightforward. Typically, females lay oval-shaped eggs either in soil or plant tissues, either singly or in groups. These eggs are protected by an external layer, an inner lipid layer, and a true shell. The first moult occurs within the egg, and the second-stage juvenile (J2) emerges by rupturing the egg shell. These larvae may either remain at the feeding site within the host or move to new locations. Juveniles resemble adults in appearance, and the adults usually remain in the soil to feed on new roots (Maggenti and Allen, 1959). Fig. 4 illustrates the nematode life cycle. The pre-parasitic stage consists of the free-living J2, which emerges from the egg and seeks out a host. The parasitic stage begins when the nematode starts feeding on host roots. During the parasitic stages, the nematode uses a stylet located at the mouth to pierce root cells and extract nutrients from plant tissues. For root-knot nematodes, the J2 penetrate the root near the tip and move towards the apical meristematic region by intracellular migration (Holbein et al., 2019). In cyst-forming nematodes, juveniles also penetrate plant roots but eventually settle at the vascular cylinder after intracellular migration. Here, they establish syncytial-feeding sites, which develop as protoplasts from dead cells are incorporated. These syncytia serve as specialized nutrient sources for the nematodes, supporting their development and reproduction through biotrophic interactions (Zhang et al., 2017). The J2 juveniles go through three additional developmental stages before reaching adulthood. Adult males leave the roots to mate with females in the soil. Once fertilized, females produce large numbers of eggs that remain within the female body, eventually forming a

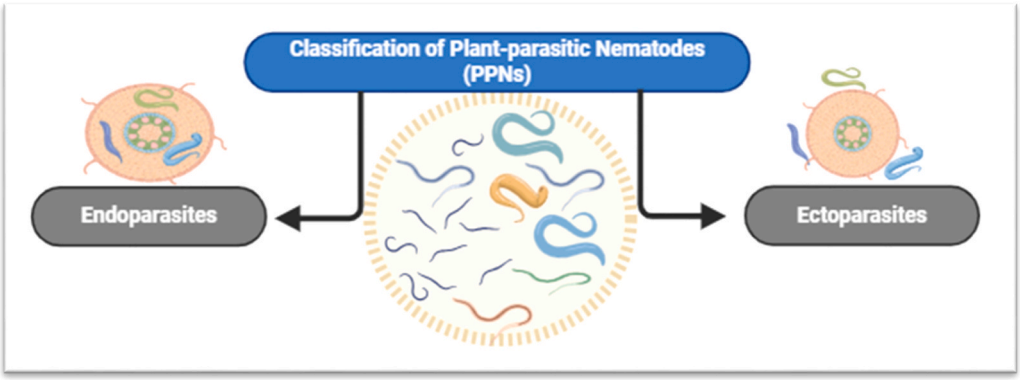

Fig. 3. Types of Nematodes based on their feeding behavior.

Table 1
Characteristics of Various Plant-Parasitic Nematodes (PPNs).

Serial no.	Nematode	Distribution	Host range	Symptoms	Life cycle duration	Feeding habit
1.	**Cyst Nematode** *(Heterodera spp.)*	North Africa, Algeria, Europe, Mediterranean	Potato, soybean, oats, wheat, barley	Stunting, wilting, chlorosis; adult female attaches to roots	Up to 55 days	Migratory endoparasite
2.	**Root Rot Nematode** *(Hrischmanniella spp.)*	Papua New Guinea, Solomon Islands	Rice, cotton, sugarcane, maize	Chlorosis, growth retardation, browning of rice roots	Up to 30 days	Migratory endoparasite
3.	**Sting Nematode** *(Belonolaimus spp.)*	Southeastern USA, Gulf of Mexico, Atlantic coast, Australia, Venezuela, Brazil	Corn, turfgrasses, peanuts, citrus, strawberry, root vegetables	Root damage, poor growth, wilting, chlorosis; can cause plant death	Up to 28 days	Ectoparasite
4.	**Citrus Nematode** *(Tylenchulus spp.)*	Worldwide in citrus-growing regions	Citrus (Rutaceae family), Poncirus trifoliata, grapes, olive, persimmon	Slow plant growth, yellow foliage, reduced fruit yield, thickened root region	4–8 weeks	Semi-endoparasite
5.	**Seed Gall Nematode** *(Anguina spp.)*	West Africa, North Africa, Australia, Brazil, China, India, Turkey, USA	Wheat, rye, barley	Leaf rolling, curling, blisters, stem distortion; high yield loss in wheat and rye	Varies by environment	Ectoparasite in early stages; later endoparasitic
6.	**Lesion Nematode** *(Pratylenchus spp.)*	Worldwide	Cereals, legumes, vegetables, fruits, coffee, peanuts	Yellowing foliage, reduced growth, root and tuber necrosis	3–9 weeks, longer under anhydrobiosis	Migratory endoparasite
7.	**Root Knot Nematode** *(Meloidogyne spp.)*	Worldwide	Cover crops, fruit trees, ornamental plants, agronomic plants	Root galls or knots, wilting, yellow leaves, nutrient deficiency	3–4 weeks at 27–30 °C	Migratory or sedentary endoparasite

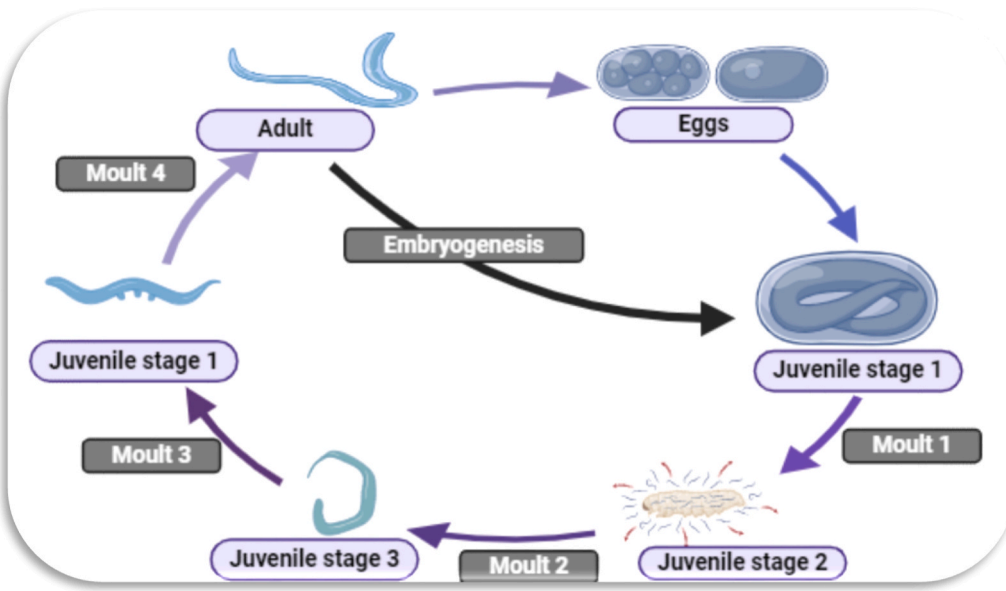

Fig. 4. Life cycle of a nematode.

protective cyst. After the female dies, eggs containing J2 larvae hatch, and the free-living juveniles will search for new roots to begin the cycle again (Jones, 1981).

4.2. The significance of exploring nematode ecology for sustainable agriculture with effective pest control

Studying the ecology of nematodes is crucial for understanding their distribution, behavior and the interactions within soil environment. Currently research in this field examines how various biotic and abiotic factors-such as temperature, moisture, soil composition, and plant species-affect nematode populations (Neher, 2010; Lazarova et al., 2021). This knowledge is valuable for developing effective strategies to manage nematode infestations, minimize the damage of crops, and close difference between the actual and the potential crop yields. By identifying that how these different factors influence nematode behavior, the researchers can create targeted management techniques that exploit nematode vulnerabilities, ultimately reducing their harmful effects on crops. Additionally, the research of nematode ecology is integral to fostering sustainable agriculture (Chaudhary and Meghvansi 2022). Nematode infestations often cause severe crop damage by feeding on plant roots, resulting in stunted growth, wilting, and decreased yields (Pulavarty et al., 2021). These infestations can lead to reductions in productivity of agricultural crops, causing financial losses to the farmers. The damage induced by nematodes also increases crop's susceptibility to other diseases, exacerbating the problem and further reducing yields (Aslam and Mukhtar, 2023a; Aslam and Mukhtar, 2023b). Nematode-related diseases pose substantial economic threats to global agriculture, with plant-parasitic nematodes (PPNs) causing approximately 12.3 % of crop losses worldwide, equating to an annual economic impact of USD 157 billion (Singh et al., 2015). In addition to direct crop damage, nematode infestations can escalate production costs

by necessitating expensive management practices, such as chemical control (Abd-Elgawad, 2021). This added financial burden is particularly challenging for farmers in developing countries with limited resources. By well understanding ecological interactions among plants, soil organisms and nematodes, researchers can help to develop the integrated pest management (IPM) strategies that reduce reliance on chemical pesticides and enhance the natural methods of pest control. Nematode infestations contribute to price fluctuations, food shortages, and diminished export revenues, all of which contributing to significant socio-economic implications. Additionally, research of nematode ecology is pivotal in advancing new technologies and tools for nematode management. For example, insights into the genetic makeup of nematodes and their interactions with plant roots can aid in the development of gene-editing techniques that impart natural resistance to nematodes in crops (Abd-Elgawad, 2022).

5. Role of nematodes towards the health of soil

Soil health refers to the capacity of soil to function within its ecosystem boundaries to support biological productivity, maintain environmental quality, and promote the health of plants and animals. A healthy soil system enables essential life processes, including decomposition, nutrient cycling, plant growth support, maintaining moisture balance and pH, fostering a dynamic soil food web, preserving microbial diversity, and controlling pollutants, pests, and pathogens. Nematodes, microscopic roundworms, play a significant role in sustaining soil health. These organisms are the most abundant multicellular life forms in soil ecosystems, with estimates suggesting that over 30 billion nematodes can exist in just one square meter of soil (Sohlenius et al., 1977). The diversity within nematode communities, including their trophic roles, contributes to various soil functions. Nematodes function across multiple trophic levels, serving as primary consumers (plant parasites), secondary consumers (bacterivores and fungivores), and tertiary consumers (predators, omnivores, and animal parasites). Plant-parasitic nematodes (PPNs), which damage plants by feeding on their sap using a stylet, are particularly significant in agriculture due to their impact on crop health (Sasser and Freckman, 1987). Some nematodes act as ectoparasites, feeding on plant roots, while others establish more intricate relationships with their hosts. In some cases, nematode life cycles align with plant growth cycles, further intensifying their effects on crops. Besides direct damage, PPNs also contribute to complex disease interactions with other pathogens. On the other hand, entomopathogenic nematodes (EPNs), which parasitize insects, benefit farmers by biologically controlling crop pests, thus reducing pesticide dependence (Koppenhöfer et al., 2020). Predatory nematodes also regulate nitrogen mineralization by consuming microbial grazing nematodes (Wardle and Yeates, 1993). While bacteria and fungi are the primary decomposers, microbivorous nematodes play a critical role in organic matter decomposition and the mineralization of carbon (C) and nitrogen (N) in soil ecosystems, both directly and indirectly. Nematodes contribute to energy transfer from plants and detritus to higher trophic levels. A presence of nematode grazers enhances plant performance in soils with complex food webs compared to simpler ones. As microbivorous nematodes graze on bacteria and fungi, they release CO_2, NH_4, and other nitrogenous compounds, thus influencing C and N mineralization (Ingham et al., 1985). Indirectly, nematodes aid nitrogen mineralization by consuming decomposer microbes, spreading microbial propagules, excreting ammonium, and stimulating microbial growth. Both bacterivorous and fungivorous nematodes promote decomposition by enhancing microbial activity, while predatory nematodes control other organisms such as nematodes, protozoa, and rotifers (Kanwar et al., 2021). In both traditional and integrated farming systems, bacterivorous and predatory nematodes contribute to nitrogen mineralization, with estimates indicating around 8–19 % of nitrogen mineralization in field settings. Bacterivorous nematodes play a more substantial role than bacteria in nitrogen mineralization due to their higher carbon-to-nitrogen ratio and lower nitrogen content (Hunt et al., 1987). Studies have also found correlations between nematode populations and the abundance of soil minerals, suggesting nematodes might influence the mineralization of multiple nutrients. In most agroecosystems, herbivores and decomposers dominate nematode communities, leading to two primary energy flow pathways. Herbivorous nematodes act as primary consumers by feeding directly on plants, while decomposers, including bacterial and fungal-feeding nematodes, contribute indirectly by consuming organisms involved in breaking down organic matter. Understanding the role of nematode communities in soil ecosystems and their impact on plant productivity is essential for managing soil health, particularly in organic farming systems that rely on ecological processes for pest and disease regulation. This knowledge is crucial for optimizing the profitability and sustainability of agricultural systems.

6. Nematodes: as bioindicators of soil health

An ideal bioindicator should reflect the current state of ecological processes, structure, and function, while also being sensitive to changes in soil conditions caused by human activities. For large-scale regional and national monitoring programs, additional considerations apply. These indicators should be suitable across various geographic regions, soil types, seasons, and vegetation. Additionally, sufficient taxonomic knowledge must be available to allow for accurate and efficient identification of these organisms. Practical considerations include minimizing the cost of sample collection and storage, and ensuring that sample processing is straightforward, rapid, and does not require frequent sampling. Building on these criteria, Kennedy and Jacoby (1999) developed a six-component framework to assess the suitability of organisms as bioindicators, which were

later condensed into four key components: 1) Demonstrated capacity to indicate ecological responses; 2) Abundance and prevalence in undisturbed environments; 3) Extensive research on the organism, ensuring easy and affordable study; and 4) Economic or commercial relevance. Using microbial communities as bioindicators of soil health presents certain logistical challenges, including temporary shifts in populations and daily variations in microbial activity. Identifying all bacteria and fungi species in a sample, particularly free-living varieties, is another significant hurdle. Soil ecosystems are home to diverse organisms, such as mesofauna (arthropods and nematodes), microfauna (protozoa and nematodes), and microbes (fungi, bacteria, and algae), all of which play critical roles in processes like nutrient cycling and respond to environmental disturbances, such as heavy metal pollution and pesticide application (Visser and Parkinson, 1992). In comparison to microbes, soil fauna offers distinct advantages as bioindicators. Since they occupy higher trophic levels in the food chain, they reflect biological, physical, and chemical soil properties linked to their food sources. Additionally, their relatively longer lifespans-ranging from days to years-provide greater temporal stability compared to often fluctuating microbial populations that respond to short-lived nutrient surges. Among mesofauna, three groups have been identified as potential bioindicators: nematodes, collembola, and mites. Nematodes, in particular, have been extensively studied for their role as indicators in both natural and managed ecosystems, largely due to their critical contributions to ecosystem processes and services, as well as the availability of more comprehensive taxonomic and functional information compared to other mesofaunal groups.

6.1. Importance of nematodes as bioindicators

Nematodes possess several characteristics that make them valuable as ecological indicators:

1. **Widespread presence and abundance**: Nematodes are found in different types of soil environments and are present in large numbers, allowing for year-round sampling in most ecosystems. This

makes them ideal for both local and large-scale monitoring (Neher et al., 1995).

2. **Ease of sampling**: Due to their small size and high population density, nematodes are relatively easy to collect and extract, reducing the costs associated with their study compared to other soil fauna.
3. **Functional diversity**: Soil nematodes represent a wide range of functional or trophic groups, displaying diverse life history strategies, such as colonizers and persisters (Bongers, 1990).
4. **Key role in the soil food web**: Nematodes occupy multiple trophic levels and are integral to the detritus-based food web. They belong to functional guilds that respond similarly to nutrient enrichment and soil disturbances (Ferris et al., 2001).
5. **Morphological differentiation**: The close correlation between nematode structure and function enables clear separation of trophic or functional groups based on their feeding structures (Yeates et al., 1993).
6. **Sensitivity to environmental disturbances**: Nematodes exhibit varying responses to physical, chemical, and biological disturbances in soil. Some species form resistant stages, such as cysts or anhydrobiotic states, which help them survive adverse conditions, while others are more vulnerable (Fiscus and Neher, 2002).
7. **Biomarker potential**: Nematodes produce heat shock proteins when exposed to stressors like heat, metal ions, or organic toxins. These proteins can be used as biomarkers for assessing soil ecotoxicology (Kammenga et al., 1998).
8. **Extensive taxonomic knowledge and analytical tools**: Nematodes benefit from well-established taxonomic knowledge and effective methods for sampling, analyzing, computing, and interpreting indices, making them more accessible than other soil organisms.
9. **Influence on soil function**: Nematode community composition directly impacts soil function, with relevance to agricultural productivity and sustainability. Nematode maturity indices can reflect successional shifts in communities due to human-induced changes.

7. Plant pathogens and the risk to the sustainability of soil

Fungi, bacteria, viruses, and nematodes are common plant pathogens that can cause significant harm to crops, thereby threatening food security (Yu et al., 2022). Of these, nematodes are especially harmful. Plant-parasitic nematodes (PPNs) are tiny worm resembling organisms which mainly live in soil, generally feed on the plant roots and in results leads to reduced yields, stunted growth and in severe cases, even the death of plant. The agricultural industry faces substantial challenges from nematodes due to the extensive damage they cause and resulting in low yield and the financial losses to farmers. Their broad range of host and their ability of persisting in soil for long periods make nematode control particularly difficult. Nematology research plays a crucial role in identifying and managing nematode species, developing new technologies, understanding nematode ecology, and equipping researchers and farmers with better tools (Pandey, 2012). Effective nematode management strategies involve the use of resistant crop varieties, cultural practices, biological control methods, and chemical treatments (Jensen et al., 2018). Emerging approaches like gene editing and the use of beneficial microbes offer potential solutions for improving resistance towards the nematode infestations. A deep understanding of ecology of nematode is essential to develop the precise and effective management strategies. Research into the nematode behavior and their interactions with the other organisms and the environmental factors provides insights into controlling their impact. Investment in the nematology research can help in promoting sustainable agriculture and enhance the global food security. Nematode infestations pose a serious threat to crops, causing extensive damage by attacking roots, which leads to stunted growth, wilting, and reduced yields (Desaeger et al., 2020). This damage significantly lowers agricultural productivity, leading to financial losses for farmers. Additionally, plants weakened by nematode damage are more susceptible to other diseases, further increasing losses. The financial impact of nematode infestations arises from direct crop damage, yield reduction, and increased production costs. Furthermore, controlling nematodes often requires expensive measures such as chemical treatments, which add financial pressure, especially on farmers in developing countries with limited resources. The effects of nematode infestations go beyond economic losses, affecting food supply, market stability, and export revenues, with broad socio-economic repercussions. Overall, nematode-related diseases result in significant annual economic losses, highlighting the urgent need for investment in nematology research to develop effective control methods, support sustainable agriculture methods, and improve global food security.

8. Nematode management research for global food security

Research in nematode management focuses on identifying the nematode species that cause crop damage, which leads to the development of effective management strategies. These strategies encompass the use of resistant crop varieties, the implementation of cultural practices, biological control methods, and chemical treatments (Saldanha et al., 2013). This research is also crucial for developing new technologies to control nematode populations, including gene-editing methods that strengthen crop resistance against infestations (Gaberthüel et al., 2022). Exploring beneficial microbes, like specific fungi and bacteria, as potential biocontrol agents for nematodes holds significant promise. Additionally, studying nematode ecology is key to deepening our understanding of their biology and interactions within ecosystems (Shah et al., 2021). The understanding is vital for crafting targeted management strategies that are suited to specific environments and crop systems. Nematology research plays a vital role in tackling the issues caused by nematode-related crop damage, which is essential for ensuring global food security and encouraging sustainable farming practices. However, the importance of this research in supporting food security worldwide is often underestimated. It serves as a foundational element in tackling the challenges associated with nematode infestations by offering a comprehensive approach to mitigate their impact on crop yields while encouraging sustainable agricultural practices (Beeman et al., 2016). By identifying the nematode species which are responsible for the crop damage, this research aids in developing customized management strategies that can effectively reduce the economic consequences of nematode infestations. Additionally, exploring the ecology of nematode provides valuable insights into their biology and ecological relationships. Understanding the complexities of nematode ecology enables researchers to develop effective management strategies that reduce economic losses and support sustainable farming practices. Nematology research strengthens the skills of researchers and farmers alike, providing training in nematode identification, management methods, and new technology applications. By equipping those involved in nematology, this research encourages sustainable agriculture and contributes to long-term food security.

9. Nematode management

A variety of methods have been developed for the control and management of plant-parasitic nematodes (PPNs), particularly those belonging to *Meloidogyne* spp., *Heterodera avenae*, *Globodera* spp., *Anguina tritici*, and *Aphelenchoides besseyi*. Some progress has also been made against other species such as *Pratylenchus thornei*, *Heterodera cajani*, and *R. reniformis*. Presently, PPN management relies on a range of practices, including chemical nematicides, bionematicides, resistant crop varieties, crop rotation, soil amendments, fallowing, flooding, solarization, tillage, and the use of certified transplants. However, since most PPNs live in the soil or within plant roots, delivering chemical nematicides directly to their immediate environment can be challenging (Chitwood, 2003). Despite this difficulty, chemical nematicides are traditionally recognized as effective methods for controlling PPNs

(Abd-Elgawad et al., 2020). Nonetheless, concerns regarding the potential risks of these chemicals to wildlife, humans, and the environment, coupled with the emergence of resistant nematode strains due to their overuse, have intensified the search for safer and more effective alternatives. Historically, nematicides such as Dichloropropane-Dichloropropene (DD) and 1,2-dibromo-3-chloropropane (DBCP) were utilized in the 1960s and 1970s in various trials against *H. avenae* and root-knot nematodes, although attempts to eradicate potato cyst nematodes in the Nilgiri hills were unsuccessful. Between the 1980s and 2020, several organophosphates and carbamates originally developed as insecticides were tested as nematicides, but most have since been banned due to health and environmental concerns. While bionematicides are generally safer alternatives, they tend to act more slowly and may not always be as effective as conventional chemicals (Abd-Elgawad and Askary, 2020). Crop rotation is a viable and safe strategy for managing PPNs; however, the absence of resistant or immune plant varieties poses a significant challenge. Recently, two new nematicides—fluensulfone (Nimitz®) from Adama and fluopyram (Velum-Prime®) from Bayer Crop Science—have emerged and are currently being tested for several crops. A key benefit of these new treatments is their requirement for minimal application rates, raising hopes for improved environmental and health safety. Considerable efforts have also been made to identify plant products and byproducts with nematicidal properties. Early on, it became clear that no single control method could sufficiently reduce nematode populations to economically acceptable levels. As a result, an integrated approach that combines multiple cultural, physical, biological, and chemical methods is necessary for maintaining low nematode densities and enhancing crop performance. Research into botanical extracts and purified secondary metabolites has garnered attention, but regulatory hurdles and lengthy approval processes have hindered their adoption (Ntalli et al., 2020). Fallowing and flooding can be used to control PPNs, although they may not be economically viable. Tillage can be effective against various pests, weeds, and pathogens; however, it may inadvertently disrupt beneficial organisms that control nematode populations, potentially increasing damage from nematodes (Timper, 2014). Certified transplants are beneficial, but care must be taken to prevent them from entering PPN-infested soils, which is often a challenging task. Hada et al. (2021) recently noted the difficulty in recommending a reliable, economical, and safe management strategy for PPNs that does not harm non-target organisms. Consequently, farmers may seek resistant varieties and management practices for PPN control, yet such resources are often limited or unsuitable for many crops. Understanding the molecular interactions between plants and nematodes could provide new insights and resources for effective nematode control. Over the last 60 years, the focus has been on developing integrated nematode management strategies that incorporate crop rotation, tillage, solarization, organic amendments, biocontrol agents, nematicidal seed treatments, and, when necessary, the application of nematicides. Nematologists must recognize that crops in the field face not only nematode challenges but also numerous other biotic stresses from insects, mites, fungi, bacteria, viruses, and weeds. Farmers are concerned with all or some of these issues alongside nematodes. Fortunately, many practices used to manage nematodes are also effective against other pests, making it easier and more profitable to adopt integrated pest management strategies that encompass both nematodes and other pests and pathogens.

9.1. Nematode control methods

The nematode control methods are: 1. Regulatory control 2. Cultural control 3. Physical control 4. Biological control, 5. Chemical control.

9.2. Regulatory control

Regulatory control of pests and diseases involves implementing legal measures to inhibit their spread, multiplication, and adverse effects. The core principle of quarantine is to prevent the introduction of nematodes into non-infested areas, thereby controlling their dissemination. Quarantine regulations are typically enforced to limit the movement of infected plant materials and contaminated soil across state and national borders. Numerous countries have developed advanced agencies to inspect and intercept plant shipments for nematodes and other pests. In certain instances, diseased or contaminated plant materials may be treated to eradicate nematodes or barred from entry entirely. Quarantine measures are vital in halting the transport of infected plants and soil to regions free of nematodes. By implementing robust quarantine practices, the goal is to safeguard agriculture and the environment from the spread and establishment of harmful nematodes and pests.

9.3. Plant quarantine in India

The Destructive Insects and Pests Act of 1914 (DIP), introduced by the Government of India, seeks to regulate the entry of foreign pests and diseases into the nation. To curb the interstate spread of pests within India, various states have enacted agricultural pest and disease acts. These regulations empower plant protection advisors or designated officials to conduct inspections and apply necessary treatments. Strict measures have been established to combat the spread of potato cyst nematode (*Globodera rostochiensis*) and red ring nematode of coconut (*Rhadinaphelenchus cocophilus*).

9.4. Cultural control methods

Cultural control methods are among the most cost-effective strategies for nematode management, requiring minimal additional expenses while being highly effective. These methods are integrated into broader agricultural practices to lower nematode populations below damaging levels. Key cultural control strategies include:

- **Crop Rotation**: By alternating non-host crops with susceptible ones, nematode populations can be significantly diminished. This approach is particularly successful when host ranges are limited and numerous economic alternatives exist, proving beneficial against nematodes like *G. rostochiensis*, *Heterodera glycines*, and *D. dipsaci*.
- **Fallowing**: Since nematodes are obligate parasites that cannot survive long without a host or moisture, allowing fields to lie fallow for certain periods can greatly reduce their populations. Generally, 6–18 months of fallowing is effective against most plant-parasitic species, leading to depletion of their energy reserves. However, the success of this practice varies and is especially advantageous against root-knot nematodes.
- **Flooding**: Saturating the soil with water reduces oxygen levels, causing nematodes to suffocate. This technique can produce harmful chemicals like fatty acids, hydrogen disulfide, and ammonia, which can be lethal to nematodes. Although not always cost-effective, it can effectively target root-knot and stunt nematodes in vegetable crops.
- **Nematode-Free Planting Material**: Utilizing nematode-free planting material is a highly effective nematode management technique that is often less costly than treating entire crop areas. This method includes using certified seeds, disinfecting infected materials, cultivating seedlings or plantlets in sterile environments, or implementing effective quarantine measures to block the entry of foreign nematode species.

9.5. Physical methods

Physical control methods for nematodes involve soil manipulation and physical disturbances to lower nematode populations. Examples include:

- **Soil Manipulation and Physical Disturbance**:

- o Researchers found that ploughing after cotton crops reduced *M. incognita* populations by approximately 40 %. Regular ploughing disrupts the nematode life cycle, resulting in reduced numbers.
- o In greenhouse studies, mixing soil containing nematodes in a cement mixer led to significant mortality due to abrasion, as only fragments of the nematodes were recovered.
- o In sandy soils, rototilling effectively reduced nematode populations. A study in The Netherlands indicated that rototilling eliminated 80 % of a *Paratrichodorus* population within the treated zone, disrupting nematode habitats.

- **Heat**:

9.6. Dry heat treatment

Dry heat can be utilised to sterilise agricultural instruments and containers as part of thermal disinfection methods. For example, treating containers to 50 °C for 30 min has demonstrated efficacy in eradicating live nematodes. This strategy enhances biocontrol strategies by minimising cross-contamination in nursery or greenhouse environments.

9.7. Flaming of metal tools

Flaming metal instruments, such shovels, sample augers or cutting tools, can be used in combination with sanitation practices to assist evacuate any nematode propagules that may be adhering to surfaces. Nematodes and other pathogens are immediately killed by this high-heat exposure, which serves as a first line of defence before biological weapons are used in the field.

9.8. Soil solarization

Soil solarisation, which builds on preventive soil management, heats the topsoil using radiant solar energy. When the wet soil is covered with sheets of transparent polyethylene for a few weeks, the top 12 in. of soil can get sufficiently warm (often over 45–50 °C) to kill nematodes and other soilborne pests. Because it lessens pest pressure and creates an environment that is more conducive to helpful bacteria, this method works especially well before applying biocontrol chemicals.

9.9. Burning of brush or leaf litter

Burning organic waste (such as crop wastes, brush or leaf litter) has been investigated for its ability to reduce pests, however it is less focused. Burning has been shown to eradicate *Meloidogyne* spp. nematodes in the top 9 cm of soil in some environments, such as Nigerian farms, though the effects are often localised and shallow. Burning is usually combined with additional techniques, such as biological treatments that further inhibit any remaining nematodes in deeper soil layers, because of its environmental limitation.

- **Steam**:
  - o Soil steaming is a common practice for managing nematodes and other pests in greenhouses but may have side effects, such as the production of toxic byproducts from organic matter in the soil and the release of harmful minerals.
  - o Soil steaming can also adversely affect beneficial soil organisms. To mitigate these impacts, aerated steam at about 70 °C is recommended, though its adoption is limited. Installing a permanent manifold system for steam delivery can ensure even penetration. Overall, while steam can be effective for nematode control, its potential side effects and implementation costs warrant careful consideration. Integrating steam treatment with other management strategies may optimize control while minimizing ecological impacts.

9.10. Biological methods

Biological control of nematodes refers to reducing their populations through the influence of living organisms, excluding nematode-resistant host plants. This reduction can occur naturally or be facilitated through environmental adjustments or the introduction of antagonistic organisms. Various biological control agents can effectively manage nematode populations, including:

- **Fungi**: Certain nematophagous fungi can parasitize and consume nematodes, helping to lower their numbers in the soil.
- **Bacteria**: Some bacteria possess nematicidal properties and can suppress nematode populations.
- **Predatory Nematodes**: Certain nematode species prey on plant-parasitic nematodes, and their introduction can aid in biological control.
- **Protozoa**: Soil-dwelling protozoa, such as amoebae and flagellates, can consume nematodes, contributing to population reduction.
- **Tardigrades**: Known as water bears, these microscopic organisms have been observed preying on nematodes in some cases.
- **Turbellarians**: Small flatworms that feed on nematodes provide a natural control mechanism.
- **Collembolans**: Also known as springtails, these arthropods can act as predators of nematodes.
- **Predatory Mites**: Some mite species target nematodes and can help regulate their populations.

9.11. Chemical control

Historically, chemical nematicides were divided into fumigants and non-fumigants based on their volatility. Fumigants are highly volatile substances that vaporize upon application to the soil, diffusing through soil pores and proving toxic to nematodes. Some earlier fumigant examples include:

- **Halogenated Hydrocarbons**:
  - o DD (a mix of dichloropropane and dichloropropene)
  - o EDB (ethylene dibromide)
  - o MBr (methyl bromide)
  - o Nemagon (DBCP - dibromochloropropane)
- **Methyl Isothiocyanate**:
  - o Includes nematicides like methan sodium and dazomet.

It is urgent to acknowledge that use of certain fumigant nematicides, particularly methyl bromide, has faced restrictions or phase-outs in various countries due to their detrimental effects on the ozone layer and potential health and environmental risks.

Non-fumigant nematicides fall into two primary chemical categories: organophosphates and carbamates. Examples of these non-fumigant nematicides, along with their trade names and formulations, exist, and they are applied by thoroughly mixing them throughout the soil profile or dispersing them in aqueous solutions. Proper application and adherence to safety protocols are critical to achieving effective control while minimizing environmental and health risks associated with these chemicals.

9.12. The reason biological control agents (BCAs) are not chemical nematicides

With the anticipated rise in food demand due to a growing global population and the pressing challenges facing our planet, agricultural practices must increasingly rely on knowledge-driven approaches to plant protection. This shift involves a more comprehensive understanding of pests and diseases (Deutsch et al., 2018; Muller et al., 2017; van Dijk et al., 2021). Synthetic pesticides have long been effective plant protection products (PPPs), significantly reducing yield losses over the

decades. However, concerns about their adverse effects on human and animal health, as well as their negative environmental impact and diminishing effectiveness due to resistance in pests and pathogens, have driven the search for new plant protection strategies. The frequent application of chemical nematicides poses environmental risks, contributing to pollution and fostering resistance in pathogens, which leads to substantial yield losses in many crops. Consequently, there is a pressing need for alternative and innovative nematicides that are eco-friendly, cost-effective and very less harmful to the host organisms (Chelinho et al., 2017). The main goal is to decrease reliance on synthetic chemicals and adopt biological methods informed by new scientific insights into the ecology, epidemiology, and molecular mechanisms underlying plant diseases, as well as a deeper understanding of plant immune systems (He *et al.*, 2021). Numerous researchers are working on developing biological nematicides derived from agents like bacteria, fungi and other microorganisms, collectively known as biocontrol agents. These agents are environmentally friendly and pose a significantly lower risk to human food chains compared to chemical nematicides (Ab Rahman *et al.*, 2018). BCAs produce bioactive compounds capable of suppressing, antagonizing, and controlling nematodes in agricultural fields (Fourie et al., 2017). They are being classified as microorganisms and the biological-based chemicals, recognized for their ability to manage various crops through biologically active compounds. Some of these organisms exhibit antagonistic properties that impact soil environments, contribute to rhizospheric ecosystems, and challenge nematodes by establishing colonies. In the European Union (EU), integrated pest management (IPM) is mandated under Directive 2009/128/EC Annex III (European Parliament and Council 2009) and is widely accepted as the standard comprehensive approach for protecting crops from harmful organisms while minimizing risks associated with PPPs to human health and the environment (Deguine et al., 2021). While IPM emphasizes 'non-chemical methods' for pest management, such as physical, mechanical, or biological control, many farmers and growers remain skeptical that alternative IPM strategies can entirely replace synthetic pesticides or effectively reduce production risks. A recent analysis of synthetic PPP uses in Europe from 2011 to 2021 showed that sales of pesticide active substances have remained relatively stable, with annual fluctuations around ± 6 % (Eurostat 2022). This stability has led to increasing societal and environmental pressure from consumers and policymakers to minimize synthetic chemicals, prioritize residue-free products, and develop new plant protection strategies informed by a better understanding of the biological mechanisms at play. Biological control methods are part of the 'intervention' toolbox within the IPM Pesticides Directive (Directive 2009/128/EC) and are regulated by the Sustainable Use Regulation (SUR). Although the SUR was rejected by the European Parliament in its initial reading (European Parliament 2023), its goals align with the EU Commission's 'Farm to Fork' and 'Biodiversity' strategies. These initiatives are integral to the European Green Deal, which aims for climate neutrality in the EU by 2050. The SUR seeks to (a) achieve a 50 % reduction in the use and risk of PPPs by 2030, relative to the average from 2015 to 2017; (b) strengthen IPM enforcement; (c) promote non-chemical alternatives to synthetic pesticides; (d) improve monitoring data availability; (e) enhance the application of regulations and policy effectiveness; and (f) support new technologies like precision agriculture. Whether the EU Commission will pursue a second reading remains uncertain. Nonetheless, the implications of this process are becoming increasingly clear, raising the question of how biological control can be positioned as a transformative element in integrated pest management. It is apparent that future reduction goals will require significant innovation in non-conventional PPPs for pest and disease control. Moreover, it is vital to prioritize the use of readily available non-synthetic methods and support their market growth through appropriate policy measures. This includes biological PPPs derived from plant compounds 'botanicals' and microorganisms, as well as the field application of beneficial organisms. Additionally, the use of biostimulants and the integration of plant–plant interactions are becoming increasingly relevant (European Commission, Directorate-General for Health and Food Safety 2020).

9.13. Biological control of plant-parasitic nematodes

The concept of "biological control" refers to utilizing animals, fungi, or other microbes to prey on, parasitize, or disrupt targeted pest species. Essentially, it involves identifying antagonistic organisms in nature that can serve as natural control agents, specifically targeting pests that are harmful to plants, animals, or humans. This approach reduces harmful organism populations by harnessing the action of their natural predators. For nematodes, biological control specifically involves decreasing nematode numbers through other living organisms—not nematode-resistant plants—via natural processes, environmental adjustments, or introducing antagonistic species. This method is sustainable and eco-friendly, reducing dependence on chemicals and supporting beneficial soil life. Recently, there's been increased interest in biological agents for controlling plant-parasitic nematodes, leading to a growing market for biopesticides and more microbial control research. These tools aim to reduce the damage of nematode infestations on crops, contributing to food security. One notable approach is using nematophagous fungi that infect and kill nematodes. These fungi can be introduced to soil as spores, where they infect and reduce nematode populations. Nematophagous fungi fall into four main categories based on their method of attacking nematodes and offer unique advantages over bacteria and viruses in nematode control. Key fungal genera effective against plant-parasitic nematodes include *Arthrobotrys*, *Duddingtonia flagrans*, *Monacrosporium*, and *Dactylaria*. These fungi have shown promising results in managing nematodes, alongside certain parasitic nematodes that can kill other nematode species. Beneficial microbes like rhizobacteria also play a role by colonizing plant roots to protect against nematodes (Castaneda-Alvarez and Aballay 2016). These bacteria boost plant growth, induce resistance, and compete with nematodes, thus lowering their numbers. Some companies, like *Pasteuria* Inc. and Koppert Biological Systems, have developed effective products targeting *Meloidogyne* spp., using microorganisms like *P. penetrans*, *P. hartismeri*, *Pochonia chlamydosporia*, *Bacillus firmus*, *P. lilacinus*, and *Trichoderma* spp., which attach to or parasitize nematode eggs, leading to nematode death (Castillo et al., 2013). Research also explores alternative biological strategies, such as using endophytes like *Fusarium oxysporum* (FO162), which can activate systemic resistance against *Meloidogyne* spp. in crops like tomatoes. FO162 colonization increases root exudates that repel *M. incognita*. A major advantage of biological control is its compatibility with other management practices, like crop rotation and resistant plant varieties, creating a holistic and effective nematode management approach. The success of biological control depends on choosing suitable agents, environmental factors, and soil health. Variability in nematode populations and their adaptability can also impact effectiveness. Ultimately, biological control is an effective and sustainable option for nematode management, reducing the need for chemicals and promoting beneficial soil organisms. Its long-term success depends on integration with other management practices.

9.14. Biological control agents (BCAs)

Some nematodes, particularly entomopathogenic nematodes, are utilized as biological control agents to manage insect pests. By decreasing pest populations, these nematodes enhance the health of plants and agricultural systems, while also lowering the reliance on chemical pesticides (John et al., 2019; Aslam and Mukhtar 2023).

- Entomopathogenic Nematodes: These parasitic nematodes have a beneficial partnership with specific bacteria in their gut. They infect insects by entering through natural openings or by piercing the cuticle. Once inside, they release bacteria that cause septicemia, leading to the insect's death within days.

- Host Specificity: Different species of these nematodes target specific pests, allowing them to focus on particular insect pests while minimizing effects on other organisms (Jansen van Rensburg, 2020).
- Effective Pest Control: Entomopathogenic nematodes are highly effective at controlling various insect pests, especially soil-dwelling larvae of beetles, weevils, caterpillars, and flies.
- Low Environmental Impact: Unlike chemical pesticides, these nematodes pose little risk to non-target organisms, including humans, pets, and beneficial insects like pollinators.
- Integration with Other Pest Management Methods: These nematodes can be combined with other pest control approaches, like cultural methods, additional biological agents, and insect-resistant plants, for a more sustainable and holistic pest management strategy.
- Soil Application: Often applied to soil, entomopathogenic nematodes locate and infect insect larvae in the soil or on plant roots, making them especially effective against soil-borne pests.
- Organic Farming Compatibility: Approved for use in organic farming, these nematodes align with sustainable practices by reducing chemical usage.
- Reduced Risk of Resistance: Due to their physical mode of action, these nematodes are less likely to lead to insect resistance compared to chemical pesticides.
- Localized Application: Entomopathogenic nematodes can be precisely applied to specific areas, reducing the need for extensive spraying and limiting environmental impact.
- Naturally Occurring: Many entomopathogenic nematode species already exist in soil ecosystems. Their populations can be boosted in pest-infested areas through controlled releases to improve pest control.

9.15. *Mechanisms of biological control*

Biological control agents (BCAs), especially bacteria and fungi, utilize various strategies to reduce populations of plant-parasitic nematodes (PPNs) in both laboratory and field settings. These agents have been shown to suppress PPNs either directly or indirectly through a variety of mechanisms (Liang et al., 2022). Biological control mechanisms can operate through antagonistic microorganisms, either directly targeting the pathogen (antagonism) or indirectly through interactions with the host. There are two main mechanisms in biological control: antagonism and induced resistance. Antagonism includes methods like antibiosis, competing for nutrients, niche exclusion, and suppression through siderophores. Induced resistance involves boosting plant defenses through systemic acquired resistance (SAR) and induced systemic resistance (ISR) against pathogens. Bacterial biological control agents (BCAs) produce various substances—such as enzymes, toxins, volatile organic compounds (VOCs) and secondary metabolites—that can directly kill the nematodes or trigger ISR in plants, which helps plants indirectly defend against the nematodes by activating defense-related genes. Plant growth-promoting rhizobacteria (PGPR) are also effective as they compete for the nutrients and the space in rhizosphere, reducing nematode numbers. Similarly, fungal BCAs produce bioactive compounds which can either kill the nematodes directly or activate plant defenses by regulating genes involved in defense. The life stages of nematodes most vulnerable to biological control are the eggs and second-stage juveniles, which exist outside the plant in the soil's water film, where they're exposed to antagonistic microorganisms that can infect and parasitize them. By targeting these stages, biological control disrupts the nematode lifecycle, lowering their population. The mechanisms by which bacterial and fungal agents control plant-parasitic nematodes (PPNs) are summarized below.

9.15.1. *Plant growth-promoting rhizobacteria (PGPR)*

Bacteria that effectively colonize plant roots are referred to as "rhizobacteria". This root colonization involves bacteria surviving on seeds, multiplying in the spermosphere due to seed exudates rich in carbohydrates and amino acids, and then attaching to root surfaces to establish themselves in the developing root system. Therefore, root colonization is an active process rather than a temporary association between bacteria and roots in the soil. The rhizosphere, which encompasses the soil surrounding plant roots, is home to numerous bacterial species that enhance plant growth by producing plant growth regulators and increasing nutrient availability; these beneficial bacteria are termed plant growth-promoting rhizobacteria (PGPR). Beyond boosting crop productivity, PGPR also stimulate a range of resistance against pests and diseases across various crops. Lugtenberg and Kamilova (2009) have categorized PGPR into three types: biofertilizers (which enhance nutrient availability), phytostimulators (which promote plant growth, often by degrading organic phytohormones), and biopesticides (which control diseases mainly through the production of antibiotics and antifungal metabolites). Nematodes and microbial communities represent two vital components of soil biotic ecosystem which have coevolved over extended periods. Using bacteria as the biological control agents in the rhizosphere of a plant can effectively suppress plant-parasitic nematodes (PPNs). Nematodes and microbial communities are two essential parts of the soil ecosystem that have coevolved over a long time. Using bacteria as biological control agents in the plant rhizosphere is an effective method to suppress plant-parasitic nematodes (PPNs) (Sun et al., 2021). Various microorganisms, including species such as *Rhizobium*, *Xanthomonas*, *Arthrobacter*, *Bacillus*, *Bradyrhizobium*, *Enterobacter*, *Frankia*, *Klebsiella*, *Proteus*, *Flavobacterium*, *Microbacterium*, *Pseudomonas*, *Serratia*, *Acinetobacter*, *Azospirillum*, *Alcaligenes*, *Agrobacterium*, *Azotobacter*, *Burkholderia*, *Cellulosimicrobium*, and *Erwinia*, are common rhizosphere inhabitants and play a vital role in root colonization. Among plant growth-promoting bacteria (PGPB), genera such as *Pseudomonas*, *Serratia*, and *Bacillus* have shown remarkable efficiency in controlling PPN populations over the past two decades (Ayaz et al., 2021). Research has focused on the mechanisms by which these bacteria reduce PPN populations, particularly by development of Induced Systemic Resistance (ISR) and the production of antibiotics during their interactions with PPNs (Siddiqui and Shaukat, 2004). In addition to primary antagonists, various rhizobacteria from specific genera are known to act against plant-parasitic nematodes (PPNs). These include *Agrobacterium*, *Arthrobacter*, *Alcaligenes*, *Azospirillum*, *Beijerinckia*, *Bradyrhizobium*, *Clostridium*, *Corynebacterium*, *Gluconobacter*, *Flavobacterium*, *Phyllobacterium*, *Phingobacterium*, *Serratia*, *Streptomyces*, *Variovorax*, *Actinomycetes*, *Aureobacterium*, *Azotobacter*, *Bacillus*, *Burkholderia*, *Chromobacterium*, *Clavibacter*, *Curtobacterium*, *Desulforibto*, *Enterobacter*, *Hydrogenophaga*, *Klebsiella*, *Methylobacterium*, *Pseudomonas*, *Rhizobium*, and *Stenotrophomonas*. Among them, *Bacillus* strains have been extensively studied for their production of the volatile organic compounds (VOCs) and secondary metabolites with the strong nematicidal effects. For example, culture filtrates from *Bacillus subtilis* OKB105 and the *Bacillus cereus* 09B18 showed up to 95 % mortality against the J2 stage of nematodes *M. javanica* and *Heterodera filipjevi* (Zhang et al., 2016). Another strain, *B. subtilis* OKB105, along with *B. amyloliquefaciens* B3, demonstrated high nematicidal activity against nematodes such as *Ditylenchus destructor, A. besseyi* and the *Bursaphelenchus xylophilus*, achieving mortality rates between 79 % and 100 %. Additionally, *Serratia plymuthica* M24T3 has been particularly effective, with 100 % mortality rate against the pinewood nematodes (Migunova and Sasanelli, 2021). While many plant growth-promoting bacteria (PGPBs) contribute to plant growth, some do not suppress nematodes, as studies in various conditions have shown. For instance, *Serratia proteamaculans* Sneb 851 has been successful against *M. incognita*, and *Bacillus thuringiensis* is known to inhibit PPNs by production of Cry proteins (Zhao et al., 2018; Senthilkumar et al., 2021). *Pseudomonas fluorescens* F113 also significantly reduces the mobility and egg hatch rate of *G. rostochiensis* J2 nematodes. Furthermore, VOCs produced by *Bacillus atrophaeus* GBSC56-like 2-undecanone (2-UD), dimethyl disulfide (DMDS), and the methyl isovalerate (MIV)-exhibit up to 90 % effectiveness against *M. incognita* by inducing oxidative stress in the

nematodes which leads to rapid mortality. These VOCs not only suppress nematodes but also promote the growth of plant and helps in stimulating the Induced Systemic Resistance (ISR) against the *M. incognita* in both greenhouse conditions as well as *in vitro* (Ayaz et al., 2021).

9.15.2. Infection by cry protein forming bacteria

B. thuringiensis (Bt) is a common spore-forming bacterium known for its production of Cry proteins, also called crystal proteins. These proteins are toxic to a variety of pests, including caterpillars, beetles, and nematodes, while being safe for vertebrates. Extensive research highlights Bt as a promising biopesticide for controlling plant-parasitic nematodes (PPNs). For instance, Prasad et al. (1972) demonstrated that applying *B. thuringiensis var. thuringiensis* significantly reduced populations of *M. incognita* (root-knot nematode) (Fig. 5). A number of Cry proteins now have been extracted from the transgenic plants and utilized globally to protect crops from nematode infestations (Li *et al.*, 2007). Currently, three families of Cry proteins—Cry5, Cry6, and Cry55-are recognized for their ability to kill nematode larvae and inhibit their growth (Luo et al., 2013). The Cry5 family includes Cry5, Cry12, Cry13, Cry14, and Cry21, while Cry6 and Cry55 consist of single proteins. Among these, Cry5B is the most researched and effective against nematodes, leading to intestinal lysis through the formation of lytic pores. Both Cry5B and Cry6Aa2 negatively affect *C. elegans*, resulting in decreased growth, reduced brood size, and impaired motility, offering an effective strategy for managing plant-parasitic nematodes (Yu et al., 2014). Researchers are exploring the genomes of various Bt strains to identify new cry toxins effective against PPNs. However, one limitation of Bt strains is their high sensitivity to solar radiation and various environmental chemicals, which restricts their application as biological control agents in agriculture.

A) ***Pseudomonas* spp.**

Certain *Pseudomonas* species including *P. aeruginosa*, *P. protegens*, *P. fluorescens* and the *P. chlororaphis*, serve as biocontrol agents capable of suppressing various soil-borne plant pathogens like oomycetes, bacteria, fungi, viruses and certain plant-parasitic nematodes (PPNs) like *Meloidogyne*. While some *Pseudomonas* species can colonize plant roots and act as opportunistic human pathogens, others are primarily saprophytic. Research has highlighted the genetic interactions between *C. elegans* and *P. aeruginosa*, which provide insights into molecular mechanisms underlying host-pathogen interactions. The strain of *Pseudomonas aeruginosa* PA-14 produces toxins that kill *C. elegans* through two primary mechanisms: a toxin-mediated fast-killing model, which involves rapid killing of nematodes within 4–24 h due to the low molecular weight phenazine toxins, and another cell growth-mediated slow-killing model, where bacteria proliferate in intestine of nematode, leading to death within 2–3 days (Tan *et al.*, 1999). A recent study indicated that pyoverdine contributes to nematode mortality by disrupting the host's iron homeostasis and triggering hypoxic responses. The "red death" phenomenon observed in *C. elegans* which was infected with the *P. aeruginosa* PAO1 is linked to iron acquisition via pyoverdine, involving the three pathways: a PhoB/phosphate signaling pathway, the pyoverdine iron acquisition system, and the MvfR-PQS pathway, which requires the formation of a red-colored PQS-Fe3 complex (Zaborin *et al.*, 2009). The quorum sensing (QS) system plays a critical role in regulating the pathogenicity of *P. aeruginosa* infections. Species of *Pseudomonas* possess the las-rhl QS system and the GacS/GacA regulatory system, both of which influence the production of virulence factors. While various genes, including LASB protease, LASA, ClpP2 protease, hydrogen cyanide, alkaline protease and antibiotic methoxy vinyl glycine, contribute to quorum control, these mechanisms can vary among *Pseudomonas* species but share common characteristics across different *P. aeruginosa* strains targeting different plant pathogens and the PPNs (Feinbaum *et al.*, 2012). Despite the effectiveness of *Pseudomonas* species as biocontrol agents against PPNs, the specifics of their interactions and the mechanisms by which they induce high mortality rates in PPNs remain unclear. *Pseudomonas* species release a variety of metabolites that exhibit significant biological activity against the PPNs. One notable metabolite, extracellular protease AprA from the *P. fluorescens* strain CHA0, has demonstrated substantial efficacy against root-knot nematodes (*M. incognita*) (Minglian et al., 2004). Additionally, other metabolites, like 2,4-diacetylphloroglucinol (2,4-DAPG) obtained from *P. fluorescens* strain CHA0, have shown the most effective biological activity against the cyst nematodes and some other PPNs (Siddiqui and Shahid Shaukat, 2003). Furthermore, genera such as *Pseudomonas*,

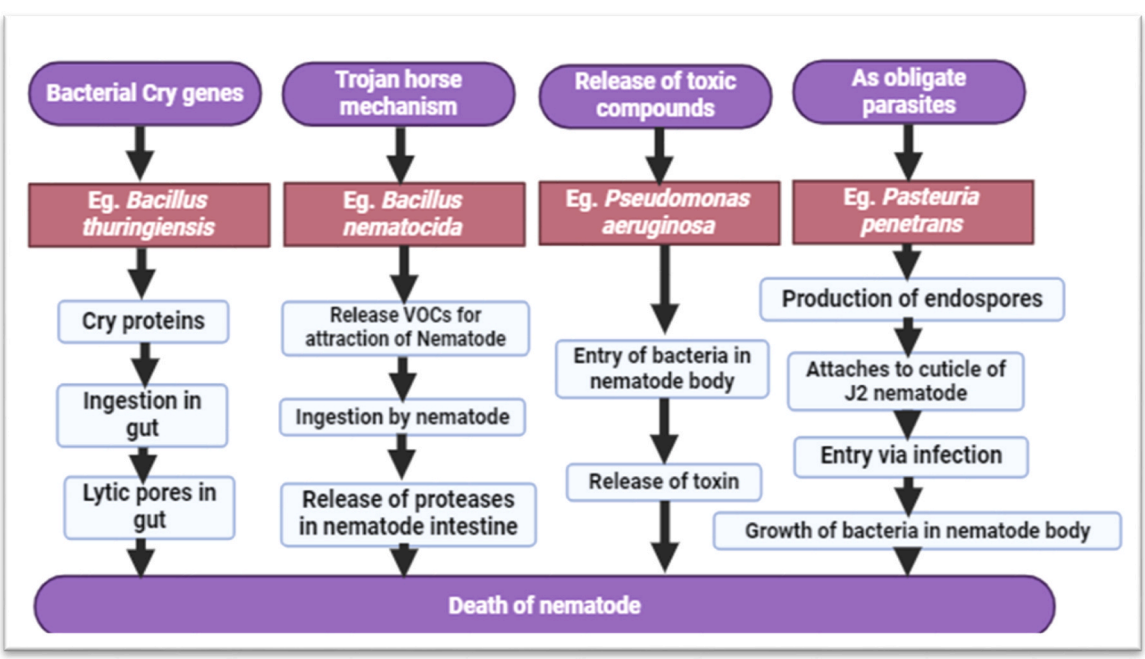

Fig. 5. Diagram illustrating pathways via which particular bacterial strains combat root-knot nematodes (RKNs), with an emphasis on extracellular enzymes and their associated molecular processes.

Bacillus and the *Pasteuria* contain large populations of the nematophagous bacteria in soil, which hold potential for controlling plant-parasitic nematode infections in both cultivated and the non-cultivated crops at a molecular level, as illustrated in Fig. 5.

9.16. Actinomycetes against Meloidogyne

According to a study done by Sun *et al.* (2006), actinomycetes, such as *P. penetrans*, are gram-positive bacteria that act as effective biocontrol agents against plant-parasitic nematodes (PPNs). *P. penetrans* produces spores in soil that attach to second juvenile stage, which is the most infectious stage and specifically targets the genus *Meloidogyne* and its larvae. After the J2 nematode penetrates the spores, they begin to germinate around eight days later, using the nematode as a food source. This actinomycete disrupts the growth and physiological processes of *Meloidogyne* species by forming a germinative tube on the nematode's cuticle, resulting in infection. Eventually, the female nematode's reproductive system is severely affected as hyphae and spores develop inside them. El-Nagdi and Youssef, (2004) reported that abamectin, a compound derived from the fermentation products of *Streptomyces avermitilis*, exhibits nematicidal properties when used in seed soaking treatments for faba beans infested with *M. incognita*. They observed that after a 24-hour exposure to 21.5 μg abamectin/mL, mortality rates reached 100 % for *M. incognita* and 97 % for *R. reniformis* (Faske and Starr, 2006). Furthermore, numerous studies have highlighted nematicidal effects of the crude extracts and then purified compounds from *Streptomyces* species against the plant-parasitic nematodes (Kaur et al., 2016).

9.16.1. Commercialization of PGPR

The commercialization process consists of several key steps, including isolating antagonistic strains, conducting screening tests, performing pot tests, assessing efficacy in the field, mass production, developing formulations, utilizing fermentation techniques, ensuring formulation viability, conducting toxicology evaluations, forming industrial partnerships, and maintaining quality control. The successful commercialization of Plant Growth-Promoting Rhizobacteria (PGPR) strains relies heavily on collaborations between scientific organizations and industry. Since rhizobia were first identified in 1886, the commercial production of Rhizobium inoculants has become widespread, especially in developed countries. In 1897, Bayer AG launched a bacteriological fertilizer based on *B. subtilis* under the brand name Alinit for use in cereal inoculation. Research in early 1950s from Russia, China, and Western countries sparked interest in utilizing microbes for plant disease management, leading to the exploration of PGPR as a viable alternative to chemical pesticides against soilborne pathogens. Due to the potential of PGPR, *B. subtilis* was first commercialized in 1985 by Gustafson, Inc. in United States, with strains A-13, GB07, and GB03 being marketed under the names Quantum, Epic and Kodiak, respectively. In the U.S., approximately 60–75 % of crops such as cotton, peanuts, soybeans, vegetables, corn and grains are now treated with the commercial *B. subtilis* products to manage soilborne pathogens. In China, PGPR has been widely applied across 48 different crops covering over 3.35 million hectares, with reported productivity increases of up to 22.5 % in potatoes and 23 % in sweet potatoes, alongside significant reductions in diseases caused by *Glomerella cingulate* and *Xanthomonas oryzae*. Among the various PGPR strains, those based on *Bacillus* have gained popularity for commercialization due to their ability of producing endospores, having ability to withstand extreme conditions like pH changes, temperature fluctuations, and exposure to fertilizers and pesticides. By the mid-1990s, *B. subtilis* began to be used for the seed dressing in the U.S., with registration across more than seven crops and applications exceeding 2 million hectares. Currently, 18 commercial products derived from *Bacillus* are employed in the China to combat soilborne diseases, covering an area of about 20 million hectares, while *B. japonicum* has been inoculated on 25 million hectares of soybeans in the South America, and about 500,000 ha of maize and wheat in Argentina and Mexico have been treated with commercial *Azospirillum* inoculants. In India, the use of PGPR has significantly increased as awareness of sustainable farming practices has grown. Over 40 stakeholders from various provinces have registered with the Tamil Nadu Agricultural University in collaboration with Central Insecticide Board for mass PGPR production. However, despite promising results in laboratory and greenhouse studies, field outcomes have been inconsistent, with variations depending on location, year, and crop type. This inconsistency is largely due to the controlled conditions of laboratory experiments compared to the multitude of biotic and abiotic factors influencing field studies. One major challenge is soil heterogeneity, which can hinder introduced bacteria from finding suitable niches. These inoculated bacteria often face stiff competition from well-adapted native microflora for nutrients and space, leading to a rapid decline in their population shortly after soil inoculation. A critical threshold of viable cells is necessary to achieve a positive plant response, typically around 10ˆ6–10ˆ7 cells per plant for *Azospirillum brasilense*, though this requirement varies among species. For PGPR to exert beneficial effects on the plant growth, they must first survive and proliferate in the microhabitats associated with root surface, competing with other microorganisms long enough to promote plant growth or protection. Factors influencing successful inoculation include (1) effectiveness of introduced bacterial strain regarding colonization, (2) competition with native microflora, and (3) the plant variety and species, as different cultivars produce varying root exudates that can affect activity of inoculated microbes and then serve as substrates for beneficial compounds. Further research on PGPR should focus on colonization, ecological interactions, survival, and the activity within plant rhizosphere. Additionally, the reproducibility of microbial inoculant effects across diverse soil types and environmental conditions needs to be rigorously tested. When utilized in the conventional agriculture, these microbes must be compatible with the chemical fertilizers and resilient against commonly used pesticides on seeds or foliage. Therefore, a comprehensive understanding of PGPR mechanisms is essential to harness their full potential in agricultural systems.

9.17. Bacterial biocontrol mechanisms for PPNs

Plant-parasitic nematodes face attack from various microorganisms found in the rhizosphere. These microorganisms significantly impact nematode populations and the dynamics of numerous antagonistic organisms and parasites in the rhizosphere. Bacteria employ different mechanisms to suppress plant-parasitic nematodes (PPNs), which helps in facilitating development of effective with good significance the biocontrol strategies against the nematode infestations. These involves both direct and indirect antagonistic type interactions between the bacteria and the PPNs. Studies show that bacteria can directly impact plant-parasitic nematodes (PPNs) by parasitism, antibiosis and colonization, which involves producing antibiotics, toxins, lytic enzymes, and volatile organic compounds. Species in the *Bacillus* genus, for instance, release cytotoxins and other harmful compounds that target nematodes. *Pasteuria* bacteria, which are parasitic to nematodes, attach to nematode cuticles and grow within them, eventually destroying their hosts (Orr *et al.*, 2020). Plant growth-promoting bacteria, like *Bacillus* spp., produce nematicidal enzymes—such as chitinases, proteases, collagenases, and lipases—that reduce PPN populations (Pires et al., 2022). For example, enzymes like glucanases, cellulases, and pectinases from *Serratia marcescens* have proven effective against *M. incognita* (Migunova and Sasanelli, 2021; Khan and Tanaka, 2023), but the cry proteins from *B. thuringiensis* can damage the internal organs of various PPNs. Furthermore, *Bacillus* spp. produce compounds like surfactin, bacteriocins, and lipopeptides to inhibit the *M. incognita* (Liang et al., 2022). Studies also highlighted how *P. fluorescens* hinders early root penetration by phytonematodes, altering root exudates like polysaccharides and amino acids to disrupt nematode behavior. The 2,4-diacetyl

phloroglucinol (DAPG) compound from the *Pseudomonas* spp. reduces the juvenile nematode movement and prevents egg hatching in potato cyst nematodes like *G. rostochiensis*. The toxins released by *Xenorhabdus* and *Photorhabdus*, found in the entomopathogenic nematodes, also suppress nematodes within plant roots and aerial parts. Meanwhile, *Pseudomonas* spp. and *Bacillus* spp. release nematicidal compounds like hydrogen cyanide and hydrogen sulfide, respectively (Pires et al., 2022). Volatiles from *B. atrophaeus*, including 2-UD and DMDS, also show strong effects against the *M. incognita* (Ali et al., 2023b). In addition, a deep-sea bacterium named *Virgibacillus dokdonensis* emits nematicidal volatiles like acetaldehyde, DMDS, 2-butanone and ethylbenzene-that act on juvenile *M. incognita* through various mechanisms like attraction, repellence, fumigation, and direct lethality (Chen et al., 2024).Indirect methods mainly employed by the PGPRs to control PPNs include nitrogen fixation, phosphate solubilization, siderophore production, induction of plant-induced systemic resistance (ISR), hormone secretion and modification of the microbiome of a plant (Ayaz et al., 2021). PGPR synthesizes indole acetic acid to boost growth of plant under the nematode pressure and induces the host tolerance against the PPNs by stimulating production of compounds such as lipopolysaccharides, siderophores, exopolysaccharides, and the N-acyl-homoserine lactones, alongside various enzymes including β-1,3-glucanase, ascorbate peroxidase, catalase, lipoxygenase, chitinase, phenylalanine ammonia-lyase, polyphenol oxidase and superoxide dismutase (Liang et al., 2022). Plant growth-promoting rhizobacteria (PGPR) are also known to colonize plant roots, which helps reduce plant-parasitic nematode (PPN) infestations. Studies have shown that *B. cereus* B1 and *P. fluorescens* P2 colonizing white clover roots can boost the plant's resistance to nematodes like *Heterodera trifolii*. In tomatoes, the combination of *P. fluorescens* CHA0 colonization and treatment of salicylic acid has been shown to trigger induced systemic resistance (ISR) against *M. javanica* (Pires et al., 2022). Endophytic bacteria within plants also offer benefits, including disease resistance and growth promotion. For example, tomato roots colonized by *B. cereus* BCM2 developed resistance to juvenile *M. incognita*, reducing nematode damage significantly (Li et al., 2023). Research on *Bacillus* species as endophytic bacteria shows that strains like *Bacillus megaterium, Bacillus licheniformis*, *Bacillus mycoides, Bacillus pumilus*, and the *B. thuringiensis* can effectively reduce number of the galls and the egg masses released by nematode *M. incognita*. *P. fluorescens* and *B. subtilis* also help banana plants resist lesion nematodes by inducing systemic resistance. Over all, diverse bacterial species provide bioactive compounds valuable for developing sustainable nematicides. Increasingly, PGPR species are recognized as effective against plant-parasitic nematodes, suppressing nematode populations by various mechanisms. Rhizobacteria like *Pseudomonas* and *Bacillus* species are especially noted for their antagonistic properties and ability to trigger ISR. This antagonism likely results from their production of hydrolytic enzymes, nutrient competition, and root colonization. Additionally, PGPR help plants manage ethylene levels in response to stress via the ACC-deaminase enzyme and produce other beneficial compounds like siderophores, antibiotics, and phytohormones. They can also degrade quorum-sensing signals and influence plant hormones like indole-3-acetic acid and cytokinin, which contributes to their protective effects against pathogens (Ayaz et al., 2021; Liang et al., 2022).

Strategy of nematophagus fungi in determining the activity of plant parasitic nematodes and their prospective role in sustainable agriculture

9.17.1. Introduction

The use of fungi as biocontrol agents to manage plant-parasitic nematodes (PPNs) has gained significant attention in recent research. Studies have demonstrated that fungi and their metabolites are effective in reducing PPN populations in key agricultural crops. Fungi are considered a rich source of bioactive compounds for PPN control, and this field is under active investigation to develop effective strategies against PPNs (Pires et al., 2022). Nematophagous fungi (NF) act as bionematicides by parasitizing nematodes, producing toxins to kill them, and inducing defense mechanisms in plants (Girardi et al., 2022). These fungi belong to diverse taxonomic groups, such as Oomycota, Zygomycota, Ascomycota, Basidiomycota, and Chytridiomycetes (Wijayawardene et al., 2020). Certain secondary metabolites produced by NF have shown strong efficacy in controlling nematodes (Seong et al., 2021). Species of *Trichoderma* are particularly recognized for their ability to parasitize nematodes, preventing them from invading plant roots, which enhances crop yield (Mukhtar et al., 2021). For instance, *Trichoderma* strain TH has been shown to inhibit root-knot nematodes (RKNs) in tomatoes by boosting levels of phenols, flavonoids, lignin, jasmonic acid (JA), and salicylic acid (SA), while reducing oxidative stress markers such as malondialdehyde (MDA) and hydrogen peroxide (H_2O_2) (Yan et al., 2021). Additionally, *Trichoderma asperellum* T00 has been reported to regulate defense enzymes and show promise against *Pratylenchus brachyurus* in soybean plants, while *Trichoderma harzianum* has demonstrated a 61.88 % reduction in *M. incognita* infections in tomatoes (Ghahremani et al., 2019). Other fungi, such as *P. chlamydosporia*, have been shown to reduce RKN infection and female reproduction in tomatoes, while *Acremonium sclerotigenum* has significantly increased juvenile mortality rates of *M. incognita*. Additionally, the new fungus named *Chaetomium ascotrichoides* was found to be effective against *B. xylophilus* with a 99 % mortality rate in pine seedlings (Kamaruzzaman et al., 2024). The Nematophagous fungi employs different strategies to target the nematodes, like predatory fungi such as *Drechslerella* spp. and *Arthrobotrys oligospora*, which usually form traps using hyphal networks, and endoparasitic fungi like *Drechmeria coniospora*, which directly parasitize nematodes. Facultative parasitic fungi, including *P. lilacinus*, *P. chlamydosporia* and *Pochonia rubescens* specifically targets eggs of nematode and adult females (Kishore et al., 2022). Some of the nematophagous fungi, such as *Catenaria anguillulae* from the Oomycetes, use adhesive spores and biflagellate zoospores to trap nematodes. Additionally, soil rich in organic matter helps these fungi thrive, with some species functioning as saprotrophs in the absence of nematodes. There are also fungi which produces toxins, like *Pleurotus ostreatus*, which immobilize the nematodes even before penetrating into their cuticles (de Freitas Soares et al., 2023).

9.17.2. Types of nematophagus fungi

Nematophagous fungi (NF) can be categorized into five distinct groups: (A) nematode-trapping fungi, (B) endoparasitic fungi, (C) egg-parasitic fungi, (D) adhesive spore-forming nematophagous fungi, and (E) arbuscular mycorrhizal fungi (AMF), as well as (F) toxin-producing fungi that enhance plant defenses or directly target plant-parasitic nematodes (PPNs) (Maia Filho et al., 2013; de Freitas et al., 2023).

A. Nematode-Trapping Fungi (NTF)/predatory fungi

NTF are soil-borne fungi that prey on other microorganisms, such as nematodes, by trapping them for sustenance. Over 200 species, primarily belonging to the phyla Basidiomycota, Ascomycota, and Mucoromycota, have been identified (Abd El-Rahman and Mohamed, 2014). These fungi employ various trapping mechanisms, including adhesive networks, knobs, constricting and non-constricting rings, to capture nematodes. Some species, like *A. oligospora*, utilize adhesive networks, while others, such as *Monacrosporium* and *Dactylaria* species, use constricting rings or adhesive knobs (de Freitas Soares et al., 2018). Though not host-specific, NTF are effective at trapping nematodes and initiating infection upon attachment. The mechanical trapping devices range from simple adhesive hyphae to more complex structures like three-dimensional adhesive networks.

B. Endoparasitic fungi

Endoparasitic fungi rely on internal infection rather than trapping mechanisms. These fungi primarily infect nematodes through spores (conidia or zoospores), which penetrate the nematode cuticle or enter through natural openings. Once inside, the spores germinate, and mycelium develops within the nematode, ultimately leading to its death. *Nematoctonus concurrent* and *Nematoctonus haptocladus* are examples of

fungi exhibiting this parasitic strategy.

C. Egg-parasitic fungi

Egg-parasitic fungi target nematode eggs, often infecting cysts or gelatinous matrices where the eggs are deposited. These fungi are particularly effective against sedentary stages of nematodes, such as *Meloidogyne* species, and include genera like *Pochonia* and *Paecilomyces*. *P. chlamydosporia* and *P. lilacinus* have shown effectiveness in controlling root-knot nematodes.

D. Adhesive spore-forming nematophagous fungi

These fungi, belonging to various classes, use adhesive spores to parasitize nematodes. For example, *Myzocytium lenticulare* (Oomycetes), *Meria coniospora* (Deuteromycetes), and *Nematoctonus leiosporus* (Basidiomycetes) are among the species that produce adhesive spores to infect nematodes. Once attached to the nematode cuticle, these spores germinate and initiate infection.

E. Arbuscular mycorrhizal fungi (AMF)

AMF are obligate root symbionts that enhance plant nutrient uptake and provide biotic and abiotic stress tolerance, including defense against plant-parasitic nematodes (Vos et al., 2012). AMF has been shown to suppress PPN populations in various plants. However, their biocontrol effectiveness depends on the environmental factors, plant species, and the AMF strain (Salvioli and Bonfante, 2013). AMF colonization with well-formed arbuscules is necessary for effective biocontrol. Some studies, such as those by Dos Anjos et al. (2010) and Alban et al. (2013), highlight the potential of AMF in nematode management, though results can vary depending on inoculation methods and plant-nematode interactions. The potential of AMF as biocontrol agents continues to be explored, particularly in combination with other agricultural practices, despite their variable effectiveness in field conditions.

F. Toxin-producing fungi

Toxin-producing fungi release substances that immobilize nematodes before their hyphae can penetrate the worm's cuticle. In vitro studies have shown that nematophagous fungi secrete a variety of compounds, which may have nematostatic or nematocidal effects. For example, *P. ostreatus* produces droplets containing a potent toxin, trans-2-decenedioic acid, which quickly paralyzes nematodes, although the exact role of these compounds in natural environments is not fully understood. Species from the *Pleurotus* and *Coprinus* genera, such as *P. ostreatus* and *Coprinus comatus*, have been identified as toxin producers. It is also possible that more members of the Basidiomycota group engage in nematophagy than previously recognized. Additionally, fungi like *D. coniospora*, *Harposporium anguillulae*, *Lecanicillium*, and *P. lilacinus* have demonstrated nematicidal and antifungal activities. *P. lilacinus* also secretes acetic acid, which paralyzes juvenile nematodes.

9.17.3. Biocontrol mechanisms of nematophagous fungi against PPNs

The use of fungi to manage populations of nematodes in crops and plants, commonly referred to as "fungal biocontrol of plant nematodes," involves different mechanisms targeting the various stages of the nematode development. Over 200 species of diverse fungi have demonstrated the ability to attack nematodes at different life stages, including juveniles, adults, and eggs (Nordbring-Hertz et al., 2006). These fungi are classified based on their strategies for controlling plant-parasitic nematodes (PPNs) (Pires et al., 2022). Nematodes and nematophagous fungi share a predator-prey relationship, where nematodes release chemical signals into the soil, which fungi can detect and respond to by developing various methods to trap or infect the nematodes. Some fungi, like *Arthrobotrys* and *Dactylella* species, prey on nematodes by using specialized structures to trap them. For example, *Arthrobotrys spp.* forms constricting rings or sticky nets to catch and feed on nematodes (Mokrini et al., 2024). Other fungi, such as *Nematophthora* and *Hirsutella*, act as parasites by invading nematodes and growing inside them, eventually leading to the nematodes' death. Certain fungi also colonize the roots of plant or rhizosphere, where they compete with the nematodes for resources, indirectly reducing the nematode populations. Some of the fungi produces different chemicals which helps in preventing nematode eggs from being hatching or interfere with their ability to locate roots. For example, *Purpureocillium lilacinum* targets nematode eggs by penetrating them with its spores (Khan and Tanaka, 2023). The Fungi like *Trichoderma spp.* release nematicidal compounds that disrupt nematode growth, reproduction, or survival. Additionally, beneficial fungi can boost plant defenses by triggering systemic resistance against nematodes. For instance, arbuscular mycorrhizal fungi (AMF) help plants absorb nutrients and water during nematode attacks while activating plant defense genes. Beneficial fungi can also stimulate salicylic acid (SA)-mediated responses, a defense mechanism similar to those plants use against harmful pathogens (Tyśkiewicz et al., 2022). In summary, fungi play a crucial role in controlling nematodes through a variety of mechanisms, including predation, parasitism, nematicidal activity, trapping, antagonism, and defense induction. These eco-friendly methods offer sustainable solutions for long-term nematode management, improves soil health and agricultural productivity while reducing the need for chemical nematicides (Lawal et al., 2022). Nematodes present barriers against fungal infections, such as their eggshell and cuticle, which vary in thickness depending on the genus (Morton et al., 2004). When nematodes come into close proximity with fungi, the latter detect chemical signals, such as ascarosides, leading to the formation of trapping structures. This adaptive mechanism helps fungi like *A. oligospora* efficiently capture nematodes only when they are nutrient-deprived and can detect their prey. Certain fungi immobilize nematodes using toxins. For instance, *Coprinus comatu* employs a spiny ball structure to immobilize the nematode *Panagrellus redivivus*, allowing the fungus to penetrate its cuticle. Endoparasitic fungi target nematodes by releasing spores that infect the nematode's tissue and later spread through escape tubes. Other fungi utilize conidia to infect nematodes internally, as seen with *H. anguillulae*, whose conidia are ingested by nematodes and penetrate their gut or esophagus. In species that target nematode eggs, the fungi form structures such as appressoria to penetrate the eggshell and consume the developing juvenile nematode. In conclusion, nematophagous fungi employ diverse mechanisms, such as parasitism, toxin production, enzyme secretion, and trapping, to control nematodes effectively (Pires et al., 2022). These strategies help reduce nematode populations and contribute to sustainable agricultural practices. Additionally, fungal biocontrol agents employ a variety of strategies to manage nematode populations, including predation, parasitism, nematicidal effects, trapping, antagonism, and the induction of plant defenses. These environmentally sustainable approaches provide effective, long-term control of nematodes in agricultural and horticultural systems, contributing to healthier soils, enhanced crop yields, and reduced dependence on chemical nematicides (Lawal et al., 2022). Nematodes possess two key structural barriers that protect them from fungal infection. The first is the eggshell, composed of three layers-the outer vitelline layer (mainly proteins), the chitin layer, and the inner lipoprotein layer-while the second is the cuticle. The thickness of these barriers differs depending on the nematode species (Morton et al., 2004). When nematodes approach fungal organisms, chemical cues are detected. Fungi may set up traps in response to these cues, essentially hunting nematodes when certain nutrients are lacking in their environment. Nematodes produce small, conserved molecules known as ascarosides, which fungi can recognize. Each nematode species produces different ascarosides, and when these are detected, fungi such as *A. oligospora* construct nematode traps. This behavior is triggered by nutrient scarcity and ensures that energy-intensive trap-building only occurs when prey is near. Some nematophagous fungi also produce toxins that immobilize nematodes. For example, the hyphae of *C. comatus* utilize spiny ball structures to attack free-living nematodes like *P. redivivus*, penetrating the cuticle and consuming the nematode's contents after immobilization (Migunova and Sasanelli, 2021). Endoparasitic fungi release spores that are attracted to nematodes and gather in their feeding tissues. Hyphae then penetrate the cuticle, and new motile spores are eventually released to infect additional nematodes. In some cases, nematodes ingest fungal conidia, which infect them

similarly. Certain fungi also target nematode eggs, using hyphae to attach to the eggshell, producing structures like appressoria to invade and consume the developing juveniles. These fungi proliferate by generating conidiophores and spreading to neighboring eggs (Pires et al., 2022). Nematophagous fungi have evolved specialized tools for attacking nematodes, functioning like sharp instruments that penetrate the nematode cuticle, causing leakage of internal fluids and allowing fungal colonization. This process generally follows a sequence where fungal hyphae expand toward the nematode, apply pressure on the cuticle, produce a penetration peg, invade the nematode's interior, and eventually exit the host (Lawal et al., 2022). Mechanisms of infection include parasitism, production of toxic compounds and enzymes, adhesion, attraction, penetration, and digestion (Pires et al., 2022).

A. Recognition: The process by which nematophagous fungi identify their prey is quite complex. While nematode-trapping fungi generally do not exhibit strict host specificity, studies on *D. coniospora*, an endoparasitic fungus, suggest a slightly higher level of specificity. However, throughout the stages of fungus-nematode interaction, cell-to-cell communication triggers specific physiological, morphological, or biochemical responses. These recognition events play a critical role in initiating the infection process (Nordbring-Hertz et al., 2006). Nematodes are attracted to the fungal mycelia, which can create traps, and to fully developed spores and traps. This process combines "close-range" or tactile communication. The subsequent steps in the infection, such as the penetration of the nematode cuticle, likely depend on host recognition.

B. Attraction: Nematodes are drawn to compounds released from fungal mycelia, trapping structures, and the spores of endoparasitic fungi. The degree of attraction depends on the fungus's form and its parasitic or saprophytic tendencies. Generally, parasitic fungi are more appealing to nematodes than their saprophytic counterparts. For instance, endoparasitic fungi that infect nematodes with conidia tend to be more successful in attracting them compared to saprophytic fungi that rely on trapping mechanisms. Volatile chemicals and exudates seem to be the primary attractants (Nordbring-Hertz et al., 2006).

C. Adhesion: Adhesion to the host is critical for fungal parasites to initiate infection. Most pathogenic and parasitic fungi adhere to their host via an extracellular matrix (ECM) or fungal sheath. In this process, a fungal lectin (a carbohydrate-binding protein) interacts with a carbohydrate receptor on the nematode. Under electron microscopy, nematodes can be observed attaching to the spores and traps of nematophagous fungi. In *A. oligospora*, for instance, extracellular fibrils surround the trap's three-dimensional net even before direct contact with the nematode. Upon contact, these fibrils align perpendicular to the host surface, facilitating fungal invasion and attachment. On the other hand, *D. coniospora* exhibits a distinct adhesive mechanism, forming radiating fibrils that attach specifically to sensory organs on the nematode's head, inhibiting nematode movement (Jansson and Lopez-Llorca, 2004). Although the exact chemical composition of the fungal surface filaments is not fully understood, they are known to contain both proteins and carbohydrate polymers.

D. Constricting Rings: Unlike other predatory fungi that use adhesive layers (nets, hyphae, or knobs) to capture nematodes, constricting rings operate through a unique entrapment mechanism. When a nematode passes through the ring, the three cells composing the ring rapidly swell inward, constricting and trapping the nematode. This reaction is triggered by the presence of the nematode but can also be induced by external stimuli, such as heat or a needle touching the inner ring surface. The ring's closure occurs in just 0.1 s and is irreversible, with a significant increase in cell volume that nearly seals the ring's opening. Once the nematode is trapped, the fungus pierces the cuticle using a penetration tube, and trophic hyphae develop inside the nematode, allowing the fungus to consume its prey. The exact mechanism behind the constriction of these rings remains unclear, but electron microscopy has revealed that the outer cell wall of the ring cells ruptures along a defined line during cell expansion. It is hypothesized that this rupture releases wall pressure, causing the elastic inner cell wall to expand after a rapid water intake. In *Arthrobotrys dactyloides*, research suggests that glycoproteins in the ring cells are activated by the pressure exerted by the nematode. This activation triggers the opening of water channels, increasing cytoplasmic calcium levels, which activates calmodulin. This sequence leads to cell expansion, constricting the ring and immobilizing the nematode (Liu et al., 2009).

E. Penetration: The attachment of the trap to the nematode trigger fungal differentiation. In *A. oligospora*, the penetration tube forms, piercing the nematode's cuticle. This process likely involves both mechanical pressure and hydrolytic enzymes that degrade the macromolecules in the cuticle, primarily composed of proteins like collagen. Various nematophagous fungi have been found to produce proteases, which play a key role in this degradation. These proteases are part of the serine protease family, closely related to subtilisin, as revealed by sequencing data. For example, a chymotrypsin-like protease is active during the penetration process in the endoparasite *D. coniospora*. Studies on subtilisin PII from *A. oligospora* suggest that this protease not only aids in penetrating the nematode's cuticle and tissue but also exhibits neurotoxic properties.

F. Nutrient digestion and storage: After penetration, the fungus begins to digest the nematode. In *A. oligospora*, the penetration tube swells into an infection bulb inside the nematode, from which trophic hyphae emerge. This stage is accompanied by significant changes in the fungal cells, with the breakdown of dense bodies in the bulb and the development of well-organized endoplasmic reticulum in the trophic hyphae. Lipid droplets accumulate within the trophic hyphae, likely serving as nutrient reserves. In contrast, *D. coniospora*, which does not form traps, lacks dense bodies and does not produce an infection bulb upon penetration. Instead, *A. oligospora* stores nutrients by producing lectin, a protein known as *A. oligospora* lectin (AOL), during nematode infection. This lectin, belonging to a family of low molecular weight proteins found in a few filamentous fungi, accumulates in the trophic hyphae and later moves to other parts of the mycelium. Although the exact role of AOL is still uncertain, it likely plays a part in the recognition of nematodes during the interaction, given that it binds to sugar structures typical of animal glycoproteins.

G. Parasitism: Nematode-parasitic fungi benefit from their hosts by extracting nutrients. This group can be classified into nematode-trapping, endoparasitic, and egg or female-parasitizing fungi. Nematode-trapping fungi develop specialized structures to capture their prey. These structures vary based on species and environmental conditions, forming adhesive nets, knobs, or constricting rings. Upon capturing a nematode, the fungus pierces its cuticle, forming a penetration peg, and the hyphae invade and grow within the nematode, eventually emerging from its body. *A. oligospora* produces a penetration tube that effectively controls nematodes, such as *M. javanica*, in tomato plants under greenhouse conditions. Endoparasitic fungi, in contrast, do not form trapping structures but use spores to infect nematodes. These fungi are often obligate parasites, completing their lifecycle inside the nematode. Spores from fungi like *Harposporium* species are ingested by nematodes, while *D. coniospora* and *Verticillium* species adhere to the cuticle and inject spores into the host. Zoospore-producing fungi, such as *Pythium caudatum*, are attracted to nematode secretions, allowing them to encyst around natural body openings before penetrating the host. Some endoparasitic fungi specifically target nematode eggs or females. The hyphae grow toward the eggs, form appressoria, and penetrate the eggshell, ultimately digesting the contents. This ability makes fungi like *T. harzianum* effective biological control agents against nematodes like *M. incognita*, significantly reducing egg masses in infected tomato plants.

H. Toxic compounds: Certain nematode-fungal (NF) species produce chemical compounds that are toxic to nematodes, resulting in paralysis, which the fungus subsequently exploits for nutrition. Some nematodes may also exhibit head shrinkage as a consequence of these toxins. These fungi are identified as toxin-producing fungi; however, research has predominantly concentrated on predatory and endoparasitic fungi (Soares et al., 2018). Most of these toxin-producing fungi

belong to the basidiomycetes group. For instance, various edible mushrooms from the *Pleurotus* genus release toxins with nematotoxic properties (Satou et al., 2008). Specifically, *P. ostreatus* produces trans-2-decenoic acid, a compound derived from linoleic acid known to be toxic to nematodes, insects, and other fungi. It is important to highlight that basidiomycetes are not the only fungi that generate such toxins; some fungi that are not nematophagous also produce compounds harmful to nematodes (Soares et al., 2018). The chemical nature of these compounds varies widely and includes simple fatty acids and other organic acids like pyrones, lactones, benzoquinones, anthraquinones, furans, alkaloids, cyclodepsipeptides, peptaibiotics, and hybrid structures such as lactam-bearing macrolactones

9.18. Enzymes

All five categories of nematode-fungal interactions-namely nematode-trapping/predatory fungi, opportunistic or ovicidal fungi, endoparasites, toxin-producing fungi, and fungi that develop specialized attack mechanisms-utilize enzymes as essential tools for infecting and digesting nematodes (Soares et al., 2018). Enzymes are macromolecules that catalyze biological reactions, enhancing their rates. Nematodes possess physical barriers, such as their cuticle, which serves to protect them against natural predators (Ekino et al., 2017). The nematode cuticle is rich in proteins, and nematode-fungi employ both mechanical and enzymatic strategies to breach this barrier. Key enzymes involved in cuticle digestion include proteases (EC 3.4), such as alkaline serine proteases and neutral serine proteases, which catalyze the hydrolysis of peptide bonds within cuticular proteins (Liang et al., 2010). For instance, alkaline serine protease produced by *Lecanicillium psalliotae* (syn. *Verticillium psalliotae*) can degrade nematode cuticles within hours and immobilize *P. redivivus* (Yang *et al.*, 2005). Additionally, *A. oligospora* produces neutral serine proteases that contribute to its pathogenicity against nematodes (Zhao *et al.*, 2004). This fungus effectively controls *Haemonchus contortus* and *C. elegans* in vitro through the production of serine proteases (Yang et al., 2022). Therefore, these enzymes play a vital role in the infection process. However, the eggs of plant-parasitic nematodes (PPNs) possess shells rich in chitin and protein. Chitinases (EC 3.2.1.14), including endochitinases and exochitinases (Tikhonov et al., 2002), catalyze the hydrolysis of glycosidic bonds between N-acetylglucosamine groups in chitin. Consequently, chitinases are crucial fungal enzymes in the infection and digestion of nematode eggs (Khan et al., 2004). *Monacrosporium thaumasium*, for example, produces extracellular chitinases that exhibit nematicidal activity against *P. redivivus*. Besides complementing mechanical mechanisms for infection and digestion, enzymes have demonstrated nematicidal effects when applied independently, without fungal presence (Braga et al., 2015). This suggests potential new strategies for controlling PPNs. Research has shown that specific enzymes like serine proteases, collagenases, and chitinases can break down the protective layers of plant-parasitic nematodes (PPNs). These enzymes help weaken the nematodes' cuticles and eggshells, making it easier for fungi to invade and establish within the nematodes.

a. Serine Proteases: Over 20 different serine proteases have been identified from various nematophagous fungi, with P32 being the first and most effective protease isolated from *P. rubescens*. This species is known to parasitize and kill nematode eggs. Numerous studies have established that these proteases serve as enzymes capable of degrading the nematode cuticle effectively. Phylogenetic analysis indicates that serine proteases can be divided into two main lineages: neutral proteases, derived from nematode-trapping fungi, and alkaline proteases, originating from nematode-parasitic fungi. These two types of proteases differ in their substrate-binding site flexibility, electrostatic surface potential, and their catalytic as well as nematicidal properties. Structural analysis of two cuticle-degrading proteases, Ver112 from *L. psalliotae* and PL646 from *P. lilacinus*, has shown that variations in substrate-binding site residues, such as S1 and S4 pockets, contribute to their differing nematicidal activities. The efficacy of fungal activity against nematodes can be enhanced through genetic modifications (Liang et al., 2010). For instance, the pathogenicity of *A. oligospora*, a nematode-trapping fungus, is increased by integrating additional copies of the endogenous cuticle-degrading protease PII into its genome.

b. Chitinases: Many egg-parasitic fungi possess chitinases that can break down the chitin present in nematode eggshells, which contain approximately 40 % (w/w) chitin. To initiate infection, egg-parasitic fungi, including *P. chlamydosporia* and *P. rubescens*, must first penetrate the nematode eggshell. To date, four chitinases from the glycosyl hydrolases family 18 (GH18) have been either cloned or purified from various nematode-parasitic fungi, including *P. lilacinus*, *L. psalliotae*, *Clonostachys rosea*, and *P. chlamydosporia*.

c. Collagenases: Collagens are important components of the nematode cuticle, and research shows that collagenases produced by nematode-infecting fungi can break down this collagen. However, collagenase enzymes are relatively uncommon in these fungi, with *A. oligospora* being one of the few that produce them. The genomes of these parasitic fungi show an expanded presence of extracellular enzymes and related genes, suggesting that these enzymes are key to their ability to infect and adapt to parasitic lifestyles. Among the expanded gene families, glycoside hydrolases (GHs) are particularly important for degrading various cellular components of PPNs, including cellulose, lignocellulose, hemicellulose, and xylans. Structural and functional analyses of carbohydrate-active enzymes have predicted the presence of 147, 226, and 305 GHs in the genomes of *Drechslerella stenobrocha*, *A. oligospora*, and *P. chlamydosporia*, respectively.

J. Special attack devices: Certain nematophagous fungi produce specialized devices for attacking nematodes. These structures function similarly to tools that nematophagous fungi utilize to damage the nematode cuticle, thereby completing the attack. The shapes of these devices vary, resembling swords, thorny rackets, or spears (Soares et al., 2018). The process of using these devices to attack can be outlined in four steps: (A) Hyphal growth is directed toward the nematode, pressing against it. (B) A penetration peg is formed to breach the nematode cuticle. (C) Following this, the infecting nematophagous fungi extend hyphae to invade the nematode's body.

9.19. Prominent nematicidal compounds from bacteria and fungi

Chemical control is a commonly used strategy in agriculture to manage nematode populations because of its quick effects. Nematicides, specialized chemical agents targeting nematodes, are especially helpful in crops where other management methods may not be practical or effective. These chemicals can be applied to the soil or directly to plant roots to reduce nematode populations and prevent crop damage. Nematicides fall into two main categories based on how they work: contact nematicides and systemic nematicides. Contact nematicides kill nematodes on direct exposure in the soil and generally have lower toxicity and shorter-lasting effects. Systemic nematicides, on the other hand, are absorbed by plant roots and spread throughout the plant, offering greater effectiveness but often with longer residual impacts and higher toxicity (Chen et al., 2020; Chen and Song, 2021). While chemical control is effective, it's crucial to use it wisely and combine it with other methods to reduce environmental harm and prevent resistance. Exploring alternative, more sustainable approaches-particularly biological control-is essential for advancing nematology and promoting sustainable farming practices. Biological controls, including those derived from bacteria and fungi, offer eco-friendly options for managing plant-parasitic nematodes (PPNs), supporting good health of soil, and promoting sustainable production of crops. The ongoing research aims to improve the effectiveness of the bacterial and the fungal nematicides, supporting the IPM and resulting in reduction of reliance on the chemically synthesized pesticides (Khan *et al.*, 2023). Here in this section important fungal and bacterial nematicides highlighted in recent studies have been discussed. The scientists Chen and Song, (2020) conducted a

review of various nematicidal compounds to analyze their structure-activity relationships and modes of action, proposing active fragments with nematicidal properties. Notably, heterocyclic compounds have been crucial in the development of novel nematicides, including fluazaindolizine, tioxazafen, fluopyram, fluensulfone, cyclobutrifluram, and imicyafos, with some still under development. For example, fluazaindolizine was registered with the Environmental Protection Agency (EPA) on September 12, 2023, as a promising agent for controlling PPNs in various horticultural crops such as tomatoes, carrots, taro, squash, potatoes, and eggplant, as well as fruit trees like citrus, grapes, almonds, and peaches. Recent innovations in seed-treatment nematicides, including pyridinyl ethylbenzamides (fluopyram), avermectins (abamectin), and phenetylarylamides (cyclobutrifluram), highlight the potential benefits of products like Tioxazafen (NemaStrike™, Bayer, Leverkusen, Germany) (Umetsu and Shirai, 2020). The use of treated seeds allows for the targeted delivery of the active nematicide ingredients in the rhizosphere, effectively addressing PPN threats to plant roots. Moreover, seed treatments can be more cost-effective compared to in-furrow applications, as they require smaller amounts of active ingredients per area. The first seed treatment utilizing these advantages, Avicta™ (abamectin), was introduced in cotton crops in the U.S. in 2006 and could potentially be adapted for other essential crops like maize and soybeans to achieve broad-spectrum nematode control (Gaberthüel *et al.*, 2022). This innovative approach has opened new market opportunities for field crops, leading many growers to opt for seed treatments over traditional in-furrow nematicides. Seeds can be treated with nematicides alone or combined with insecticides for dual purposes (Wilson and Jackson, 2013). The development of novel nematicides with improved characteristics could further close the existing gap. Tioxazafen, as a new class of nematicides, features a disubstituted oxadiazole structure, allowing it to persist in plant roots for approximately 75 days during crop growth while selectively inhibiting nematode energy production (Umetsu and Shirai, 2020). Fluopyram has demonstrated high efficacy against several significant PPN genera, including root-lesion nematodes, root-knot nematodes (RKNs), cyst nematodes, and potato rot nematodes. A two-year field study aimed at controlling *M. incognita* in watermelon explored the efficacy of three novel nematicides (fluensulfone, fluazaindolizine, and fluopyram), applied at various times (before transplanting, at transplanting, and after transplanting) and in different combinations, including with an older product, oxamyl. Results indicated that all nematicide treatments significantly reduced *M. incognita* galls on watermelon roots compared to untreated controls, with most treatments also leading to lower nematode populations in the soil (Nnamdi and Hajihassani, 2023). In summary, the authors Nnamdi and Hajihassani, (2023) concluded that utilizing drip irrigation for the application of the novel 3-F nematicides presents a cost-effective approach for managing *M. incognita* in watermelon cultivation. A key focus in the development of these chemicals is their unique nematicidal functions and modes of action, prompting further investigation into how these new nematicides combat nematodes. For example, fluensulfone appears to gradually inflict metabolic damage that depletes lipid reserves, ultimately leading to the death of plant-parasitic nematodes (PPNs) (Chen et al., 2020). Conversely, fluopyram functions as a succinate dehydrogenase inhibitor, disrupting mitochondrial complex II in the electron transport chain (Burns et al., 2015). This mechanism differs significantly from traditional nematicides, such as organophosphates and carbamates, which inhibit acetylcholinesterase. However, a potential drawback of fluensulfone is its toxicity to non-target organisms, despite its versatility in application methods, including drip irrigation and foliar spraying. Chen et al. (2020) noted that fluensulfone's effects on PPNs can vary based on soil type and application technique. Additionally, a newer mode of action involving tioxazafen has been investigated, targeting ribosomal function (You et al., 2023). A noteworthy study published in *Nature* (Burns et al., 2023) has opened new pathways for nematode management by identifying three novel imidazothiazole-based compounds-selectivin-A, selectivin-B, and selectivin-C-that exhibit significant nematicidal properties. These selectivins were effective not only against *M. incognita* but also against four species of free-living nematodes and a cattle nematode parasite, with minimal toxicity to non-nematode organisms, such as human cells, fungi, fish, and insects. This suggests that selectivins may offer greater specificity towards nematodes compared to existing commercial nematicides. The research found that the action of selectivins is linked to microsomal cytochrome P450 (CYP) enzymes, which are involved in activating dormant pro-drugs into active agents (Ortiz de Montellano, 2013). Importantly, the metabolites derived from selectivin-A are largely absent or significantly reduced in non-nematode organisms, indicating a high level of specificity in their action. The study specifically identified *M. incognita* as possessing the CYP enzyme CYP4731A3, which bioactivates selectivin-A, leading to nematode damage. Moreover, the pursuit of new strategies for managing PPNs in key crops has led to the development of bionematicides-biological control agents (BCAs) or their bioactive compounds. These products generally avoid the negative side effects associated with chemical nematicides, offering effective control against resistant pathotypes while being safe for beneficial organisms and free from harmful residues (Abd-Elgawad, 2022). The demand for reliable and environmentally safe nematicidal options has driven the production of several bionematicides (D'Addabbo et al., 2023; Catani et al., 2023). Wilson and Jackson (2013) reviewed the active ingredients of four prominent bionematicide products produced by leading international companies: Nortica® (from *B. firmus*), DiTera® DF and DiTera wettable powder (WP) derived from *Myrothecium verrucaria* fermentation, Econem™ (from *Pasteuria usgae*), and MeloCon® or BioAct WP (from *Purpureocillium lilacinus*). Specific strains of *B. firmus* can colonize the egg sacs of *Meloidogyne* spp., effectively destroying the eggs and leading to reduced populations of both nematode galls and infectious juveniles (Keren-Zur et al., 2000). The cell-free filtrates of these strains produce toxic metabolites that can cause paralysis and death in larval and adult nematodes, including *M. incognita*, *Radopholus similis*, and *D. dipsaci* (Mendoza et al., 2008). As for the *M. verrucaria* fermentation products, their efficacy is believed to stem from a combination of active components rather than a single toxin. Wilson and Jackson (2013) summarized the mechanisms of action, noting that these products kill PPNs upon contact, inhibit egg hatching, disrupt nematode movement and host-finding capabilities, and alter the rhizosphere microbiota to enhance microbial antagonism. Currently, there are seven known species of *Pasteuria* that target nematodes, though it is likely that many more taxa exhibit significant intra-species diversity (Abd-Elgawad, 2024). Among these, at least four species are utilized in control programs for plant-parasitic nematodes (PPNs): *P. penetrans* for *Meloidogyne* spp., *P. thornei* for *Pratylenchus* spp., *P. nishizawae* for both *Heterodera* spp. and *Globodera* spp., and *P. usgae* for *Belonolaimus* spp. (Gaberthüel *et al.*, 2022). As there are no commercially available products containing *Pasteuria thornei* endospores, Stirling (Abd-Elgawad, 2024) recently recommended a method to acquire these endospores by adapting the widely used dried root technique for *P. penetrans*. This involves culturing *Pratylenchus* species in vitro on sterilized carrot tissue, where target nematodes are added to an endospore suspension derived from spore-filled cadavers and inoculated with *P. thornei* spores. Following this, the nematodes are introduced to a host plant, with roots harvested after 6–8 weeks, air-dried, and ground into a powder. However, if the nematodes remain alive within the roots, air-dried soil should be utilized as an inoculum source for innovative integrated nematode management (INM) strategies (Abd-Elgawad, and Askary, 2018). *P. lilacinus*, marketed as a water suspension, can be applied through irrigation or with ground application equipment. The water-dispersible granule (WG) formulation contains 1 × 10^10 spores, while the wettable powder (WP) formulation has 1 × 10^11 fungal spores per gram of product. While the application methods, timings, rates, targeted PPN species, and host plants for these emerging biocontrol agents (BCAs) are often detailed on product labels (Wilson and Jackson, 2013; Abd-Elgawad, 2024), some

are still in development or expected to be available soon. Additional tables feature fungal and bacterial BCAs used with other components in INM, along with their commercial products, formulation types, and mechanisms of action against PPN-infested crops (Abd-Elgaw et al., 2018). Some commercial bionematicides also incorporate plant-derived ingredients. For example, *NEMguard*® is marketed as a bionematicide based on garlic extract. Its active compounds, including allicin and polysulfides, provide nematicidal, insecticidal, and fungicidal effects (Umetsu and Shirai, 2020). *NEMguard WF* has been approved for use on various crops such as carrots, potatoes, and tomatoes, and is under evaluation for additional applications on fruit trees like grapes (D'Addabbo et al., 2023). Recent findings indicate that this product may serve as an effective and safe management solution for dagger nematodes (*Xiphinema index*) in organic and integrated vineyard systems, suggesting that it and similar bionematicides could become widely accessible in the near future. Diallyl polysulfides, the active constituents of *NEMguard*®, appear to exhibit multi-site activity, as proposed by Chatterji et al. (2005). They observed DNA damage and/or cellular apoptosis in the tested PPNs following the production of reactive oxygen species from polysulfide compounds. Additionally, disturbances in metal homeostasis were suggested due to the metal-binding properties of diallyl polysulfides (Anwar et al., 2014).

9.20. Bacterial-based nematicides

Research has shown that *Bacillus* and *Pseudomonas* species, along with other bacteria in the rhizosphere, can produce antimicrobial compounds harmful to nematodes (Ali et al., 2023). A significant amount of investigation has centered on the lytic enzymes produced by rhizobacteria as a means of managing plant-parasitic nematodes (PPNs). These bacteria synthesize various lytic enzymes, including chitinase, protease, cellulase, lipase, and keratinase, which disrupt the biological processes of nematodes by degrading crucial components of their exoskeleton and eggshell (Migunova and Sasanelli, 2021). The enzyme chitinases break downs chitin in nematode eggshells, causing them to rupture and allowing immature eggs to hatch. Proteases, on the other hand, break down the peptide bonds in glucose-protein complexes, weakening nematode's structural integrity (Kishore et al., 2022). An enzyme protease recently isolated from *B. cereus* NJSZ-13 has shown effective nematicidal action against the pinewood nematode. When the collagen in nematode cuticles is broken down, the nematodes become more vulnerable to predatory fungi and antagonistic bacteria. For example, collagenases from *Brevibacterium frigoritolerans* FB37BR and *P. fluorescens* FP805PU have demonstrated effectiveness against *Meloidogyne ethiopica* and *X. index* (Li et al., 2023). Additionally, the lipid layer in eggshells of cyst nematodes like *Heterodera* provides protection against drying out. However, *B. thuringiensis* strains FB833T and FS213P, along with *Bacillus amyloliquefaciens* FR203A, have strong lipase activity that disrupts this layer, showing potential as nematicides against *X. index* (Castaneda-Alvarez *et al.*, 2016). Other enzymes like glucanases and pectinases from the *Pseudomonas* spp., affect soil populations of *M. incognita*. These enzymes work alongside the secondary metabolites for controlling plant-parasitic nematodes (PPNs), suggesting a need for more research on their nematicidal properties (Kishore et al., 2022; Ahmed et al., 2023). In addition to the lytic enzymes, *B. thuringiensis* produces various Cry proteins with significant nematicidal effects on PPNs. Various Cry proteins, including Cry12, Cry13, Cry5, Cry6, Cry14, Cry21 and Cry55, damage nematode intestines, disrupting cellular and vacuole functions (Liang et al., 2022). Other bacterial metabolites like sphingosine from *B. cereus* cause oxidative damage to *M. incognita* (Kishore et al., 2022). The *B. thuringiensis* also produces thuringiensin which is thermostable compound that acts against soybean cyst nematodes (*H. glycines*) by mimicking ATP and inhibiting RNA synthesis, similar to the insecticide aldicarb. Thuringiensin-producing strains also make trans-aconitic acid, an inhibitor of tricarboxylic acid (TCA) cycle enzyme aconitase (Seong et al., 2021). Volatile compounds like 2-nonanone and 2-undecanone from *Bacillus* spp. have shown nematicidal activity against pinewood nematodes, with 2-nonanone specifically damaging the nematode's pharynx and intestine (Deng et al., 2022). The *P. fluorescens* produces 2,4-diacetylphloroglucinol (2, 4-DAPG), which disrupts mitochondrial membrane function and inhibits *M. incognita* egg hatching, though it doesn't affect juvenile nematodes. Other metabolites, like hydrogen cyanide from some *Pseudomonas* strains, inhibit mitochondrial cytochrome oxidase, helping control PPNs like *M. javanica*. Actinomycetes, being prokaryotic in nature, are known for releasing different variety of antimicrobial metabolites that target different pathogens (Pires et al., 2022). For example, avermectin B1 (abamectin) which is a macrocyclic lactone from *S. avermitilis*, is used to manage insect and parasitic infections by targeting glutamate-gated chloride channels, showing high toxicity to PPNs like *M. incognita* and *B. xylophilus*. Similarly, milbemectin from the *Streptomyces hygroscopicus* acts through a similar mechanism. Spectinabilin from *S. spectinabilis* has shown potent effects against *Bacillus xylophilus*, with LC50 of about 0.84 mg/L. The Spinosad, mixture of spinosyns obtained from the *Saccharopolyspora spinosa*, targets *M. incognita* by affecting GABA and nicotinic acetylcholine receptors, leading to paralysis (Kishore et al., 2022). Despite of these findings, the detailed actions of many microbial metabolites remain unclear, limiting our understanding of their bioactivity against PPNs. Techniques like omics and molecular docking could help in studying these compounds, enabling more targeted and sustainable management of PPNs in agriculture.

9.21. Fungal-based nematicides

Fungal biocontrol agents are emerging as a promising source of new nematicides. Certain fungi, including *Nematoctonus*, *Arthrobotrys*, *Pochonia*, *Pleurotus* and the *Trichoderma*, have demonstrated significant nematicidal properties through release of toxins, volatile organic compounds (VOCs), enzymes and other secondary metabolites (Pires et al., 2022). Compounds from these fungi show considerable effects against plant-parasitic nematodes (PPNs). Eg. a compound Aurovertin D from *P. chlamydosporia* has proven highly toxic to *M. incognita*, with an LC50 of 16.45 μg/mL (Wang et al., 2015). The Phomalactone, a secondary metabolite produced from *Nigrospora* species, significantly reduced the invasion of *M. incognita* J2s at 500 mg/L in the tomato roots (Kishore et al., 2022). Other compounds like Thermolide from *Talaromyces thermophilia* effectively target both the *M. incognita* and *B. xylophilus*, with the LC50 values between 0.7 and 1.0 μg/mL. A compound Hypocrellin A produced from *Shiraia bambusicola* also has a notable effect, causing 50 % mortality in the *B. xylophilus* at an amount 50 μg/mL (Seong et al., 2021). Additionally, the 3-hydroxypropionic acid (3-HP) from the endophytic fungus named *Phomosis phaseoli* demonstrates selective action against the *M. incognita*, with an LC50 of about 12.5–15 μg/mL. A compound Flavipin from the *Chaetomium globosum* disrupts nematode energy production and protein synthesis, making it effective against soybean cyst and the RKNs (Kishore et al., 2022). Other active metabolites, like fusaric acid and bikaverin from the *Fusarium oxysporum*, inhibit *B. xylophilus*, with LC50 values of about 50 μg/mL and 43 μg/mL, respectively, potentially through mechanisms like metal ion binding. Beauvericin, a depsipeptide from *Beauveria bassiana*, targets *M. incognita* by inducing oxidative stress and activating mitochondrial cell death pathways (Youssef et al., 2020). A metabolite Enniatin B which is a cyclodepsipeptide, acts as ionophore, showing toxicity against *M. javanica*, while a metabolite Emodin, a quinone from the *Aspergillus glaucus*, has strong effects on *M. incognita*. Overall, advancing fungal biocontrol research, supported by the omics and the biotechnological tools, holds great potential for developing eco-friendly nematicides to manage PPNs sustainably in agriculture.

10. Conclusion

In summary, microbial agents, especially nematicidal compounds

from Plant Growth-Promoting Rhizobacteria (PGPR) and nematophagous fungi, offer a promising path toward more sustainable agriculture. These microbes help maintain soil health by balancing biological interactions that naturally suppress nematodes and pests, which in turn strengthens crop resilience and boosts productivity. With advancements in molecular biology and biotechnology, we have new tools for developing effective biocontrol strategies, though there are still obstacles to overcome. Some biocontrol products are commercially available, but large-scale use remains limited due to inconsistent field results. Improving fermentation and formulation processes is essential to produce reliable and effective PGPR inoculants. To encourage farmer adoption, there needs to be a greater focus on product testing, education, and effective marketing. While biocontrol agents are an environmentally friendly alternative to chemical nematicides, more field research is needed to confirm their effectiveness and consistency. Nematophagous fungi are particularly beneficial in soils with high nematode presence, providing a natural and sustainable pest control solution across various farming systems. Their ability to manage harmful nematodes without impacting other organisms highlights their value in integrated pest management (IPM). However, broader adoption depends on continued research, real-world testing, and clear demonstrations of their effectiveness to farmers. As agriculture shifts towards sustainability, microbial biocontrol agents like PGPR and nematophagous fungi are expected to play a major role in future farming, supporting eco-friendly crop protection and healthier soils.

Future prospects

The integration of microbial biocontrol agents, such as nematophagous fungi and bacteria, offers promising advancements for sustainable pest management across horticulture, environmental, and biotechnological sectors. Continued research will expand our understanding of the mechanisms and interactions between these microbes and plant-parasitic nematodes, enabling the development of optimized biocontrol solutions tailored to horticultural and agricultural needs. Biotechnology innovations, including genetic engineering and multi-omics approaches, are expected to enhance the efficacy and adaptability of these microbes. Additionally, precision agriculture tools will facilitate targeted applications, maximizing effectiveness while minimizing environmental impact. Successful adoption will require collaboration among researchers, agricultural extension services, and farmers, with comprehensive training programs to equip farmers with the skills to apply microbial biocontrol agents effectively. Adapting regulatory frameworks to expedite approval and commercialization will also be essential, encouraging market growth in line with rising consumer demand for sustainably produced food. Overall, prioritizing research, collaboration, and education will allow horticulture and agriculture to leverage microbial biocontrol, supporting resilient ecosystems, improving biodiversity, and contributing to sustainable food systems.

CRediT authorship contribution statement

Swati Meel: Writing – review & editing, Writing – original draft, Visualization, Software, Data curation. **Baljeet Singh Saharan:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

The authors declare that no generative AI or AI-assisted technologies were used in the writing, editing, or content generation of this manuscript. All content is solely the result of the authors' original research and intellectual effort.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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